Wired 04 | Operations, Dashboards, and Events

Overview

In this lab, we'll explore CloudVision's operational tools for managing and monitoring your campus network — from topology visualization and compliance enforcement to custom dashboards and event notifications.

What you'll learn:

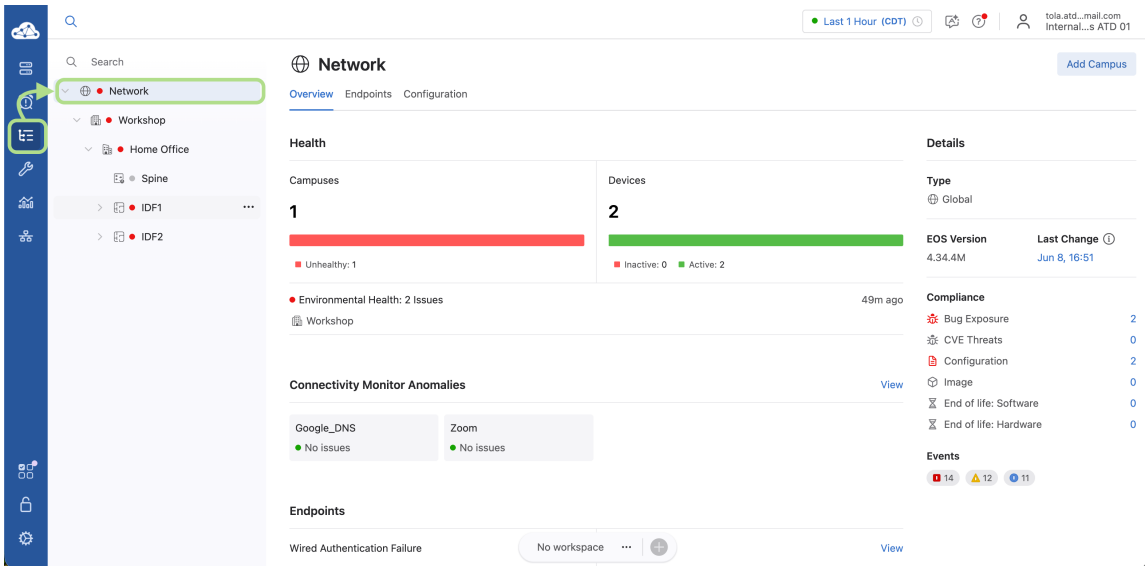
- Navigate the Network Hierarchy to inspect campus topology
- Identify and resolve configuration compliance violations
- Build a custom dashboard with device and interface metrics
- Filter, acknowledge, and export events for troubleshooting
- Search and inspect endpoints with Endpoint Overview
- Configure email notifications for critical network events

Open Launchpad | **Login Help** | **Lab Topology**

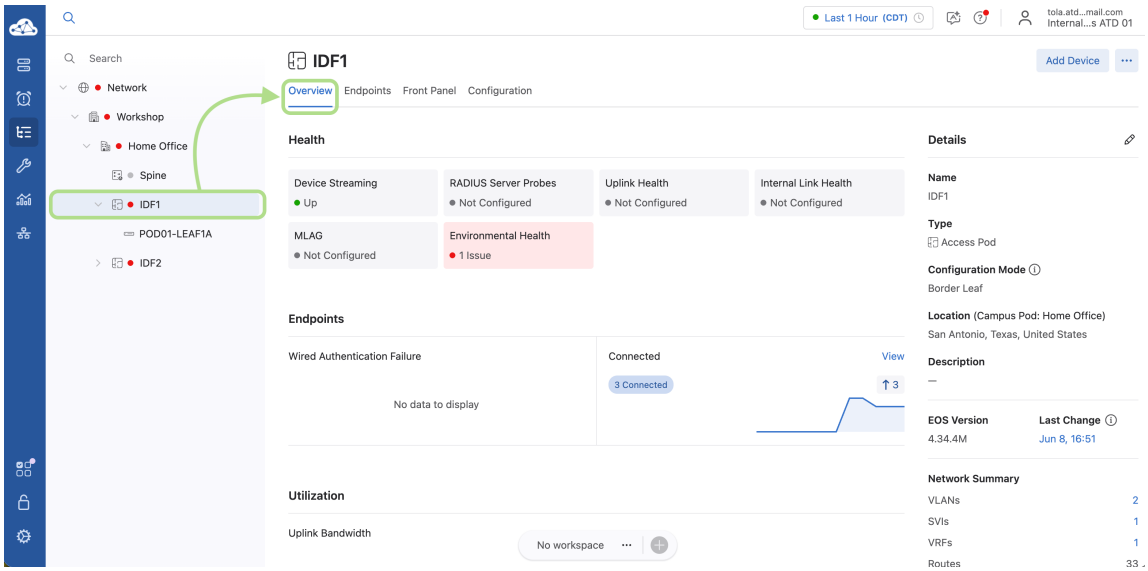
Intro to Network Hierarchy

Campus networks can be extensive, spanning many sites, buildings, and floors. The Network Hierarchy view gives you a structured way to navigate your deployment — and it aligns closely with how you'll manage your wireless access points in CV-CUE.

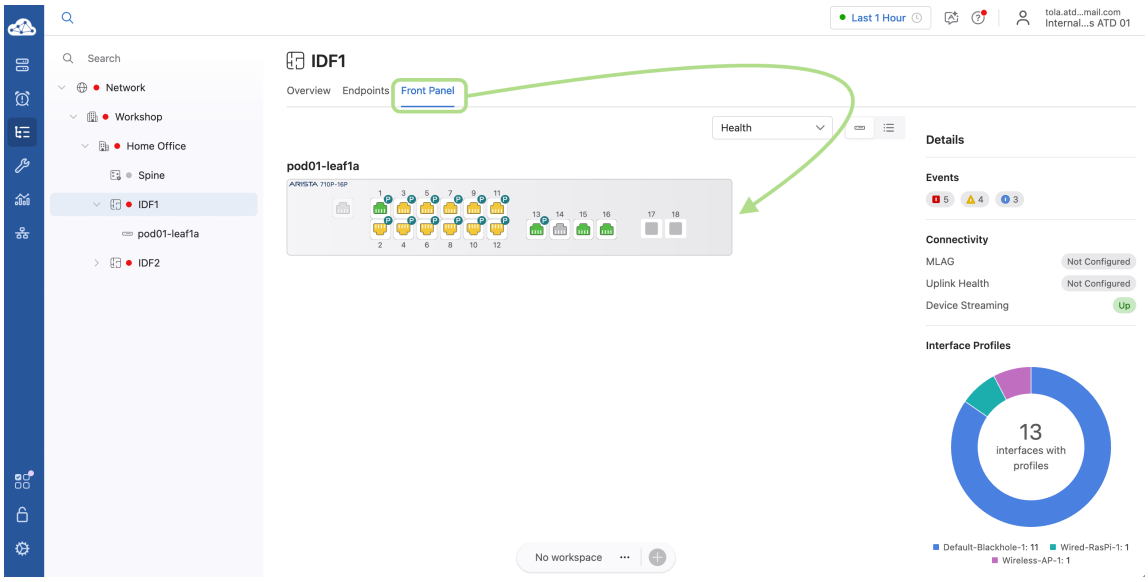
1. Select the `Network Hierarchy` tab on the left, then select `Network`. Note that your `Network` contains the workshop campus, campus pod, and access pods as you drill down.



2. Select `Workshop` to view a summary of all devices deployed under the top-level campus.
3. Select `IDF1` or `IDF2` where your device is located. Note that we now get more detail about the switches and clients connected to those devices in the `Overview` tab.



4. Select the `Front Panel` tab to view more detail about what's connected, the capabilities of your switch, and access to quick actions for interface changes.



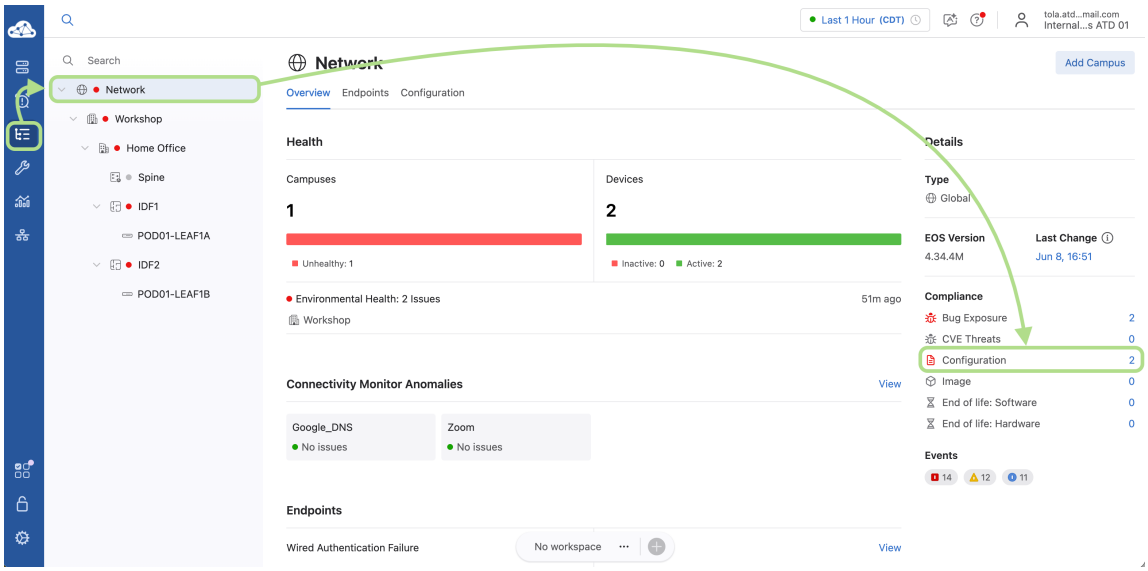
Now let's use Network Hierarchy to drill down into a compliance issue.

Configuration Compliance

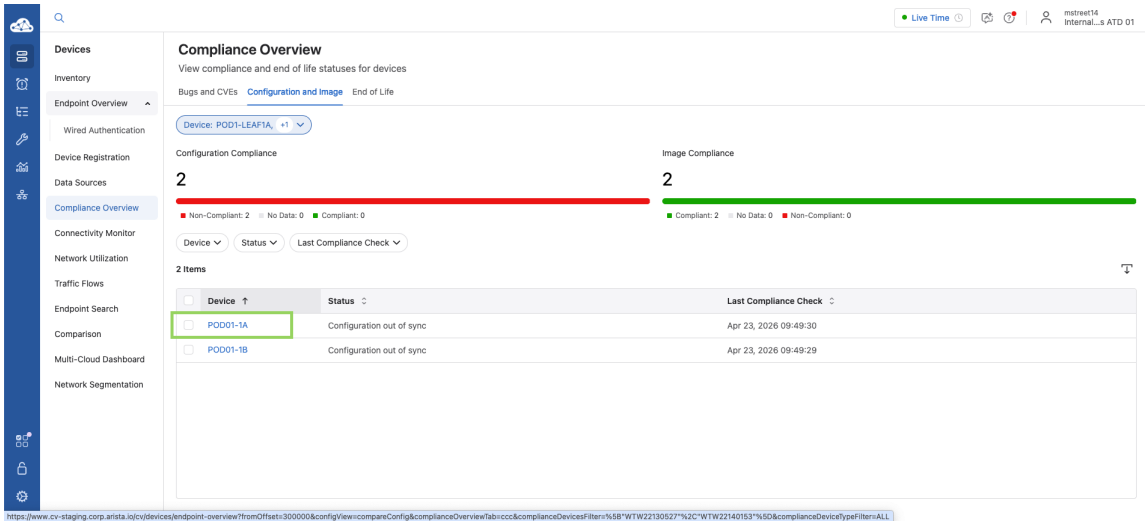
The Network Hierarchy view is one of many ways to visualize your campus — allowing you to drill down to a particular area of the network to configure or troubleshoot an issue.

We'll use the compliance panel to identify devices that are flagged due to the rollback performed earlier. This will open the compliance dashboard where we can sync the config via Change Control.

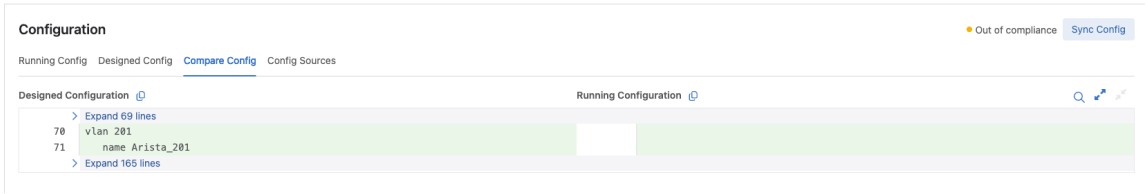
1. If you're not already on the `Network Hierarchy` page, select the tab on the left.
2. Select the top-level `Network` object, and take note of the `Compliance` panel on the right. You should see 2 devices violating the `Configuration` compliance item.



3. Select `Configuration` in the `Compliance` section. You should see the `Compliance Overview` with our 2 devices out of configuration compliance.



4. Select one of the devices to inspect what portion of the configuration is not in compliance.



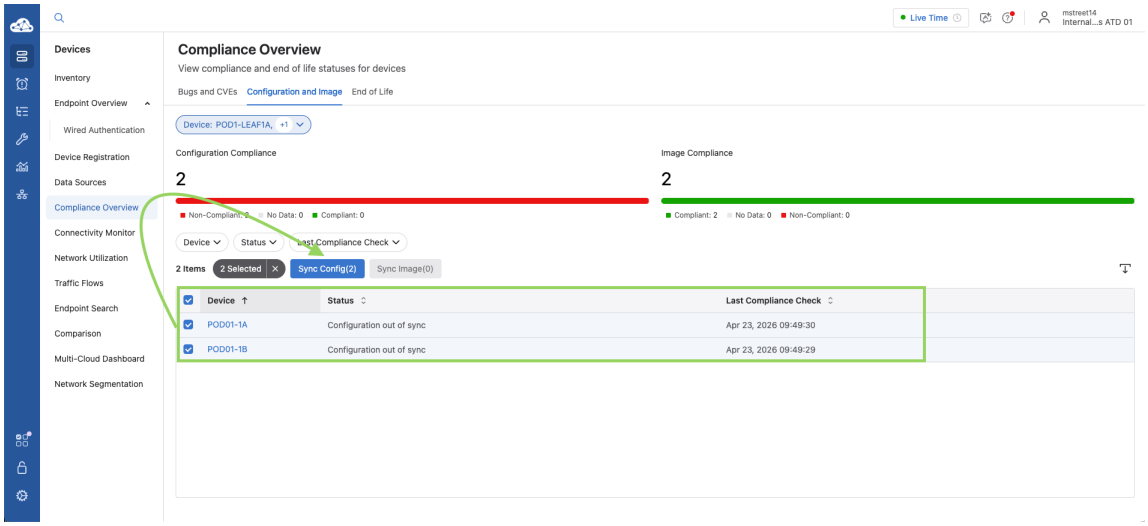
5. Review the designed configuration against the running configuration. Based on the difference, we need to revert to the designed config. Select the back button in your browser to return to the compliance page.

Why is our configuration out of sync?!

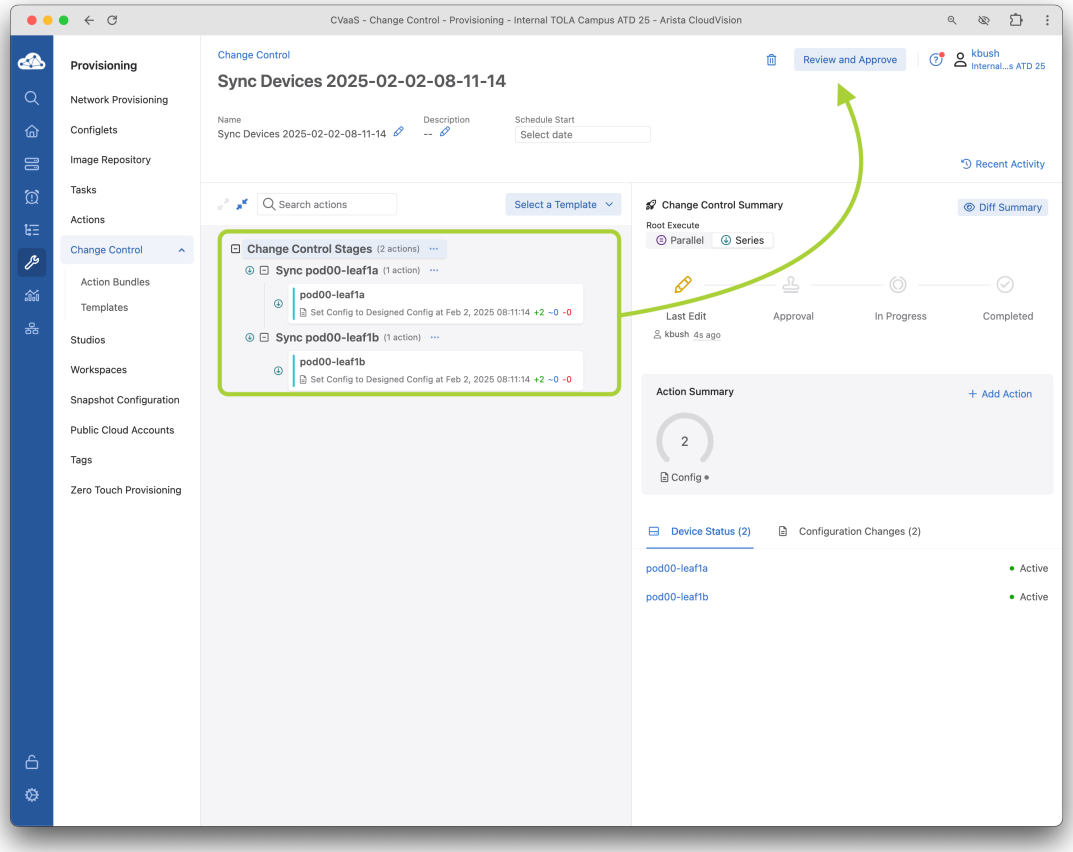
Recall that CloudVision acts as the "Source of Truth" for your configuration. It contains a designed configuration it considers the standard, driven through Studios. When we added our VLAN in the previous lab and rolled that change back, a discrepancy was created.

This could be a real-life scenario where a VLAN was added, rolled back due to an unforeseen issue, and scheduled to be added at a later date. We need that VLAN, we're reminded the device is out of compliance — and possibly for a good reason!

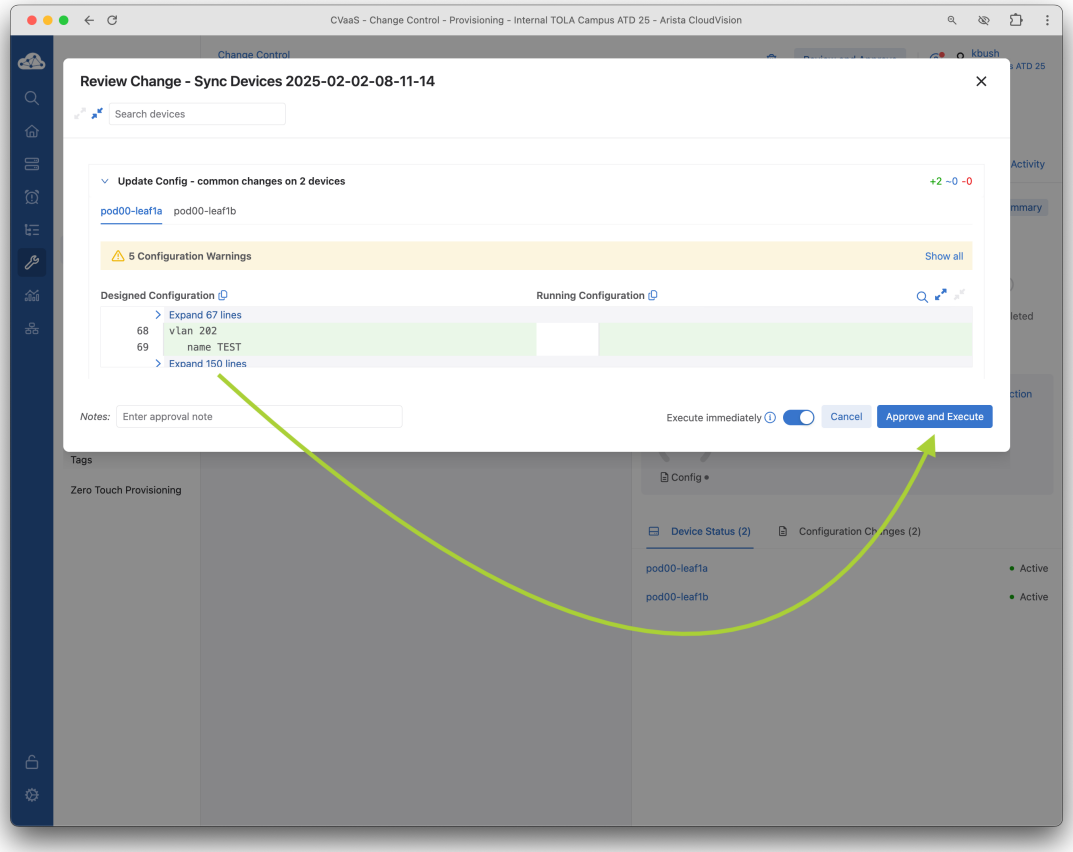
6. Select both devices and select `Sync Config`. This should be done by only one student in the pod!



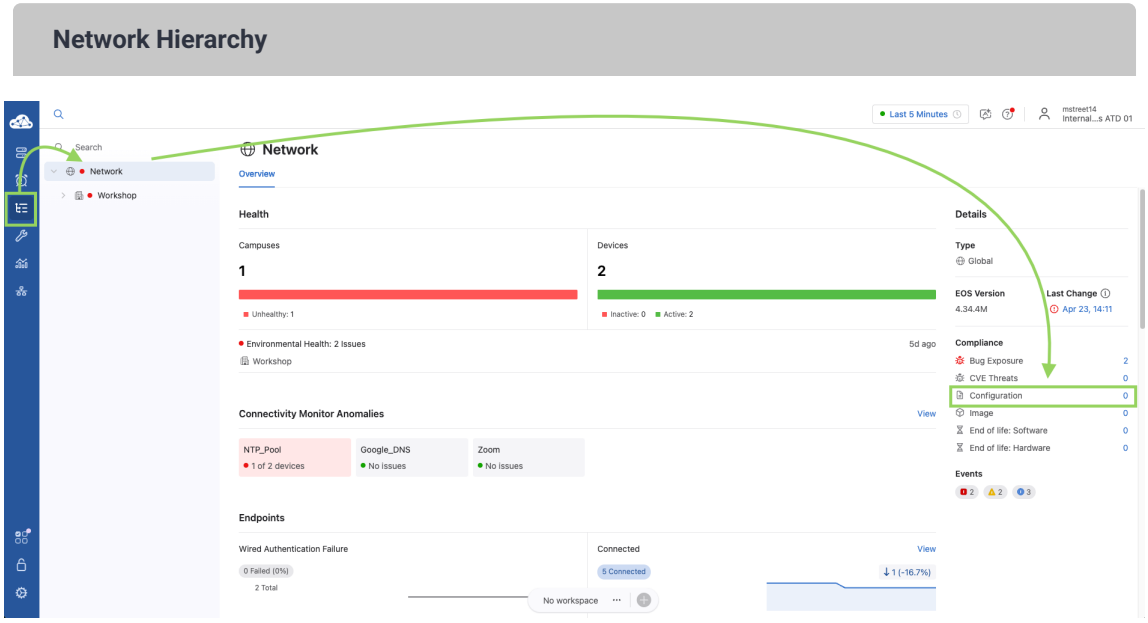
This will create a Change Control with our two devices.

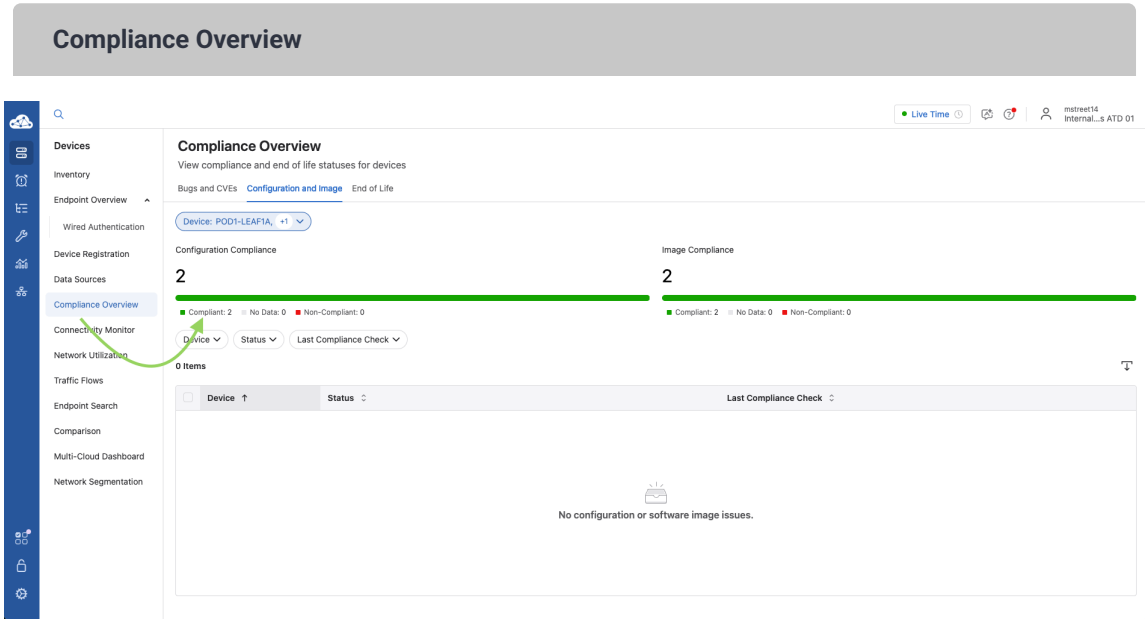


7. Select `Review and Approve`.
8. Select `Approve and Execute` when you're ready.



9. CloudVision is pushing the designed configuration back to the devices. Navigate back to `Network Hierarchy` or `Devices > Compliance Overview` — you should see your switches in your pod are now in compliance.

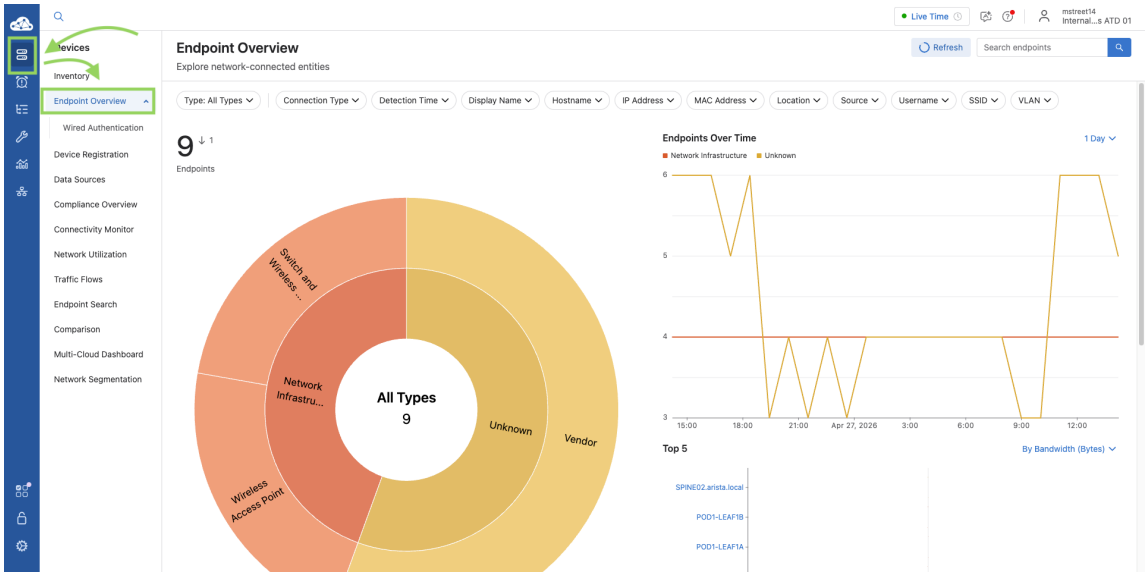




Endpoint Overview

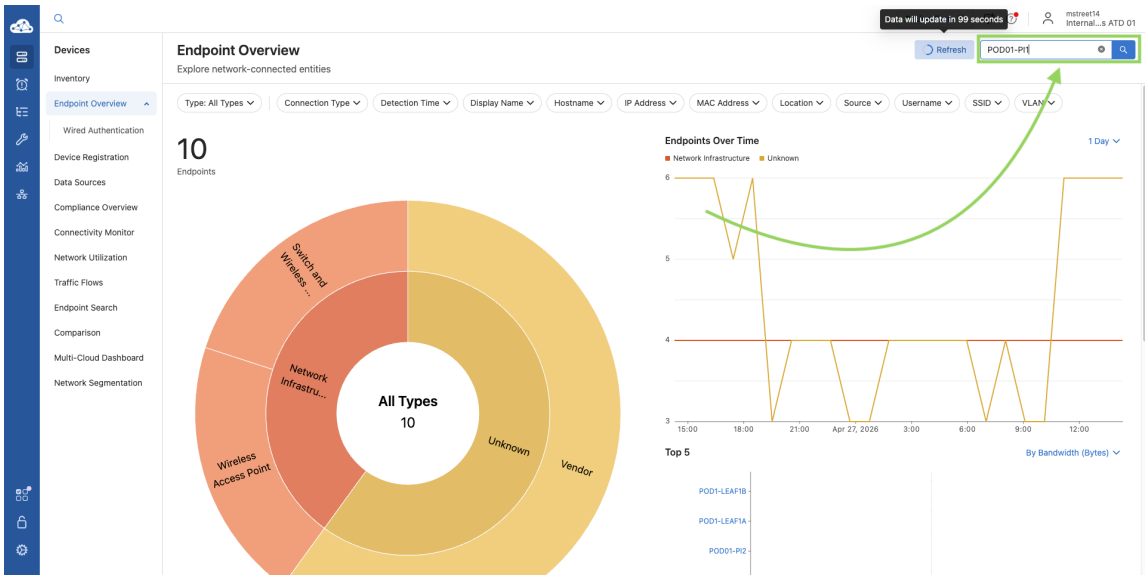
Troubleshooting client issues can be time-consuming. Endpoint Overview provides a way to search for where a client is connected and view specific information about that client. It maintains client history and shows how the client is connected, with links to further troubleshoot connection issues.

1. Select `Devices`, then `Endpoint Overview` on the left.



The Endpoint page contains high-level information about the number of clients and client types, with a list of endpoints near the bottom.

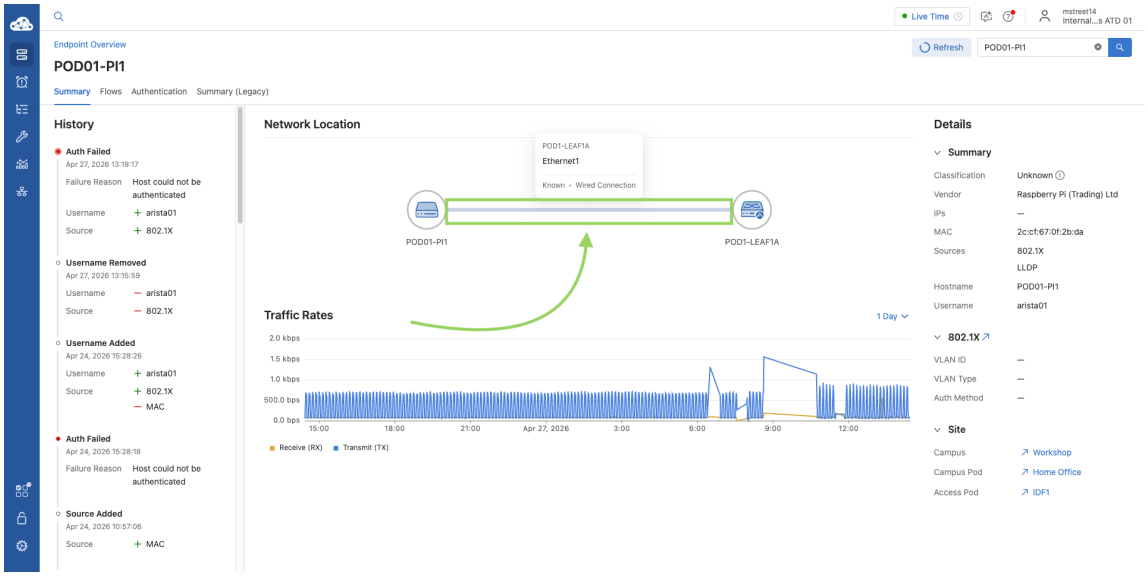
2. Let's search for an endpoint. There are two ways to search: using the filter dropdowns or the search bar. We'll use the search bar. In the search endpoint bar, enter `pod##-pi1` or `pod##-pi2` and select the magnifying glass.

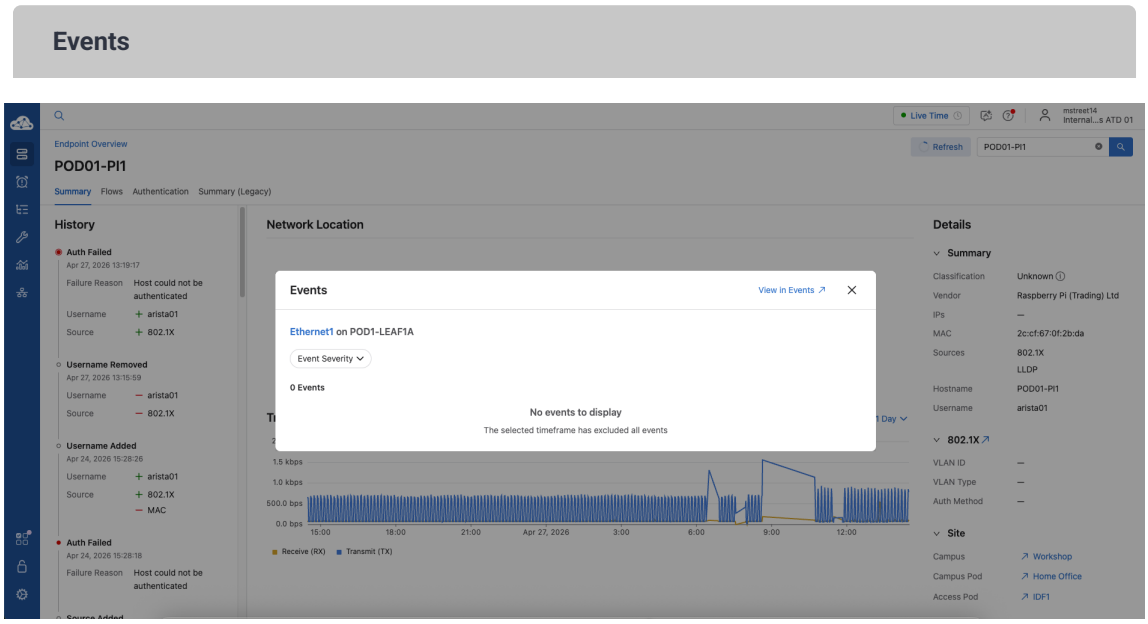


The Endpoint screen shows detailed information about the client, including client history, client details, and the network location. Take a moment to review each area.

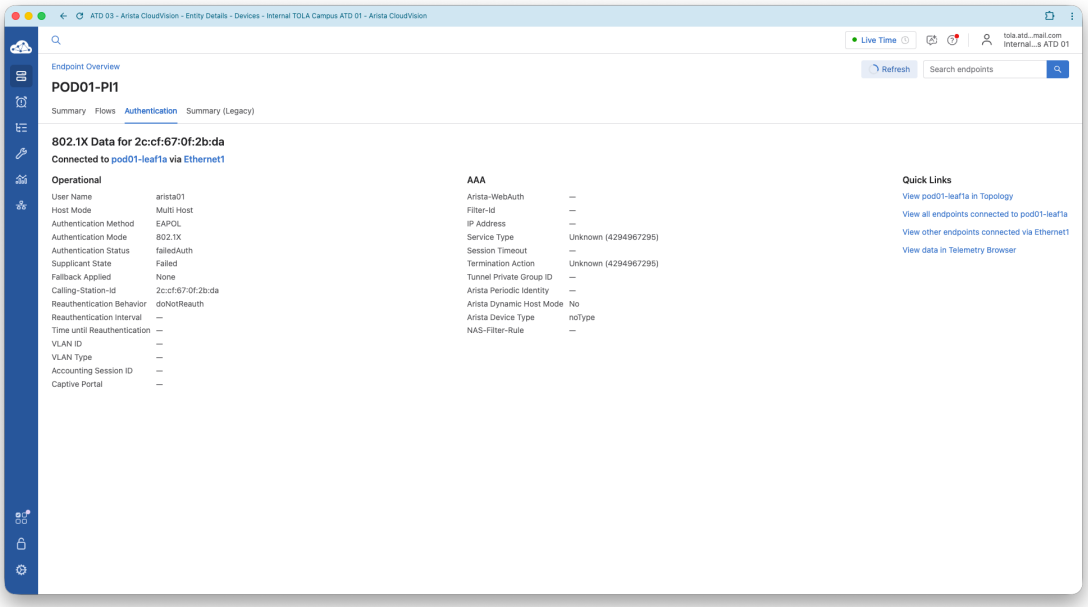
3. Hover over the link between the device and the switch in the Network Location area, then select the link to view events associated with the connection.

Hover





4. Select the `Authentication` tab to view more detailed information about how the client authenticated.

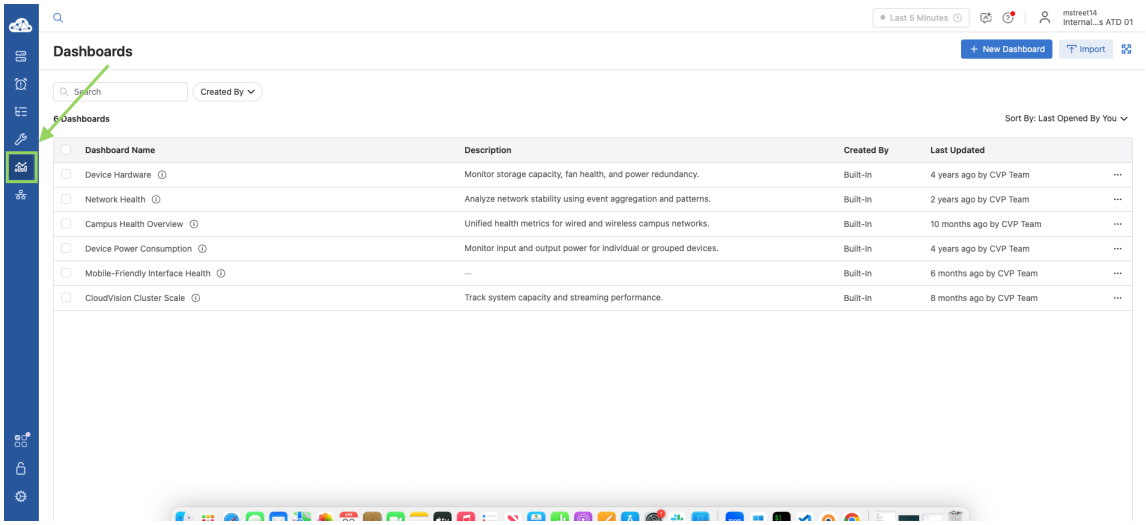


Dashboards

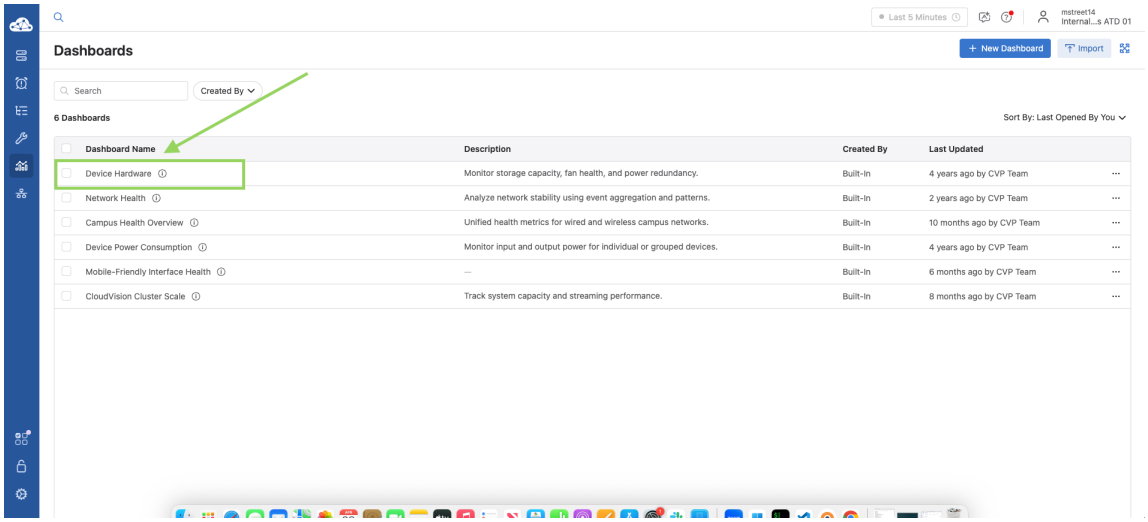
Dashboards are an important way to visualize commonly requested information. This section shows you how to navigate the built-in dashboards and customize your own.

Device Hardware

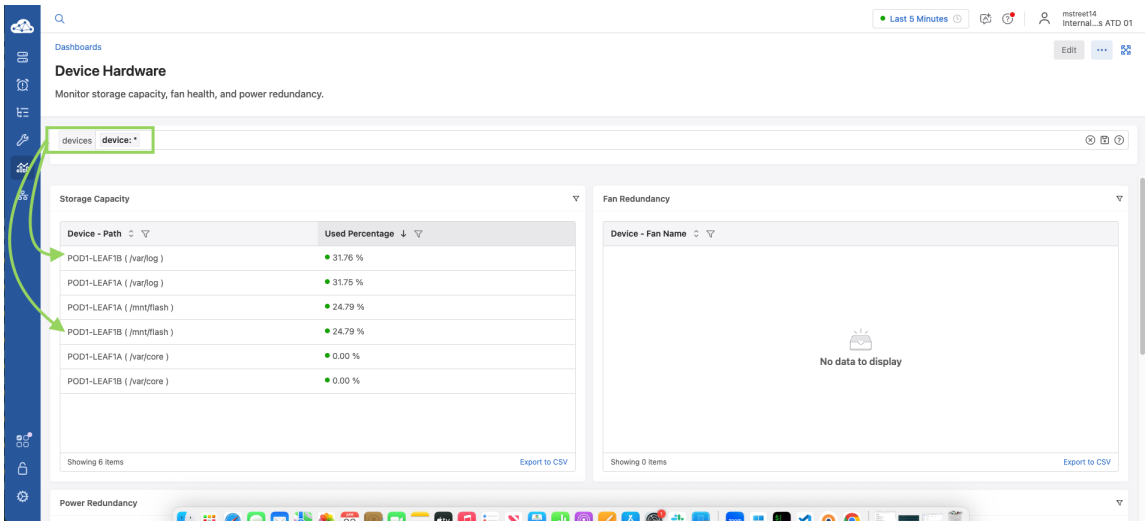
1. Open the `Dashboards` section. You'll see the built-in dashboards ready to use.



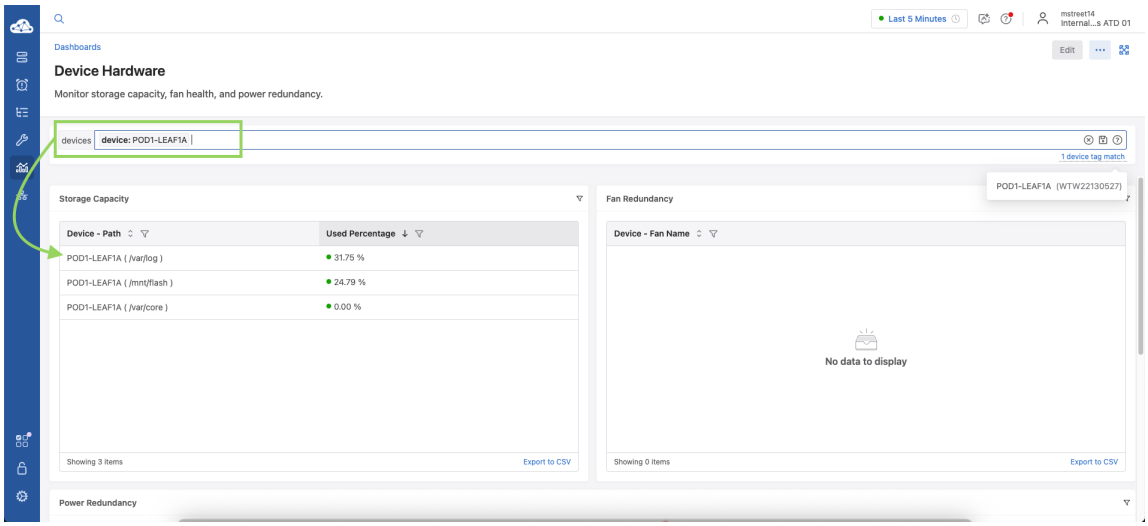
2. Select the `Device Hardware` dashboard.



By default, the dashboard selects all devices.



3. Change the dashboard to select only `leaf1a` or `leaf1b` by changing from `device: *` to `device:` to match a single device. The dashboard will filter to results for only that device.



4. Navigate back to the `Dashboards` landing page by selecting `Dashboards` in the upper left.

Custom Dashboard

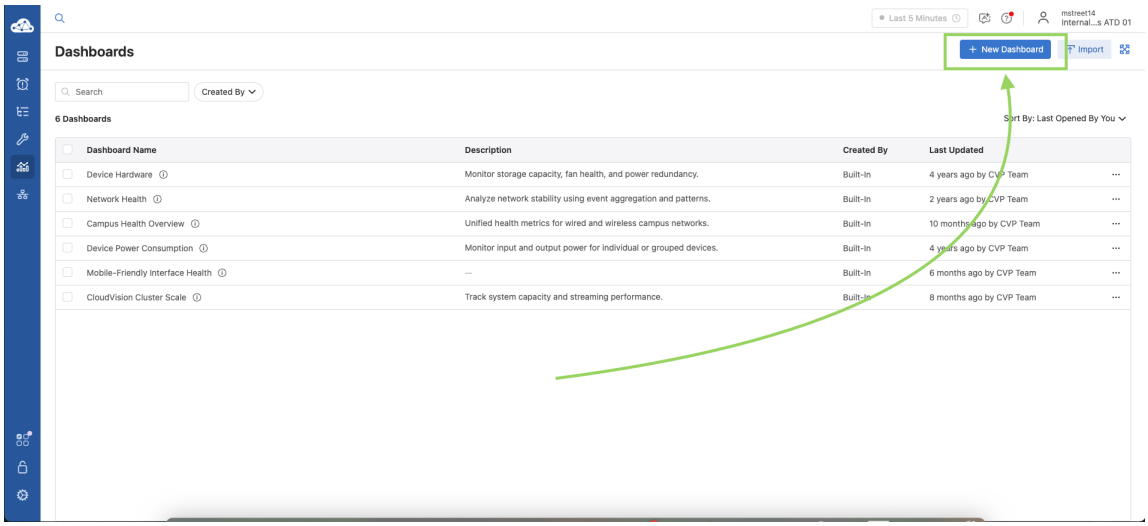
Next, let's add a new customized dashboard. There are three main constructs we will touch on here:

- **Metrics (Devices)**: telemetry data specific to a device
- **Metrics (Interfaces)**: telemetry data specific to the interfaces of a device
- **Summaries**: a metric or set of metrics summarized into a single view

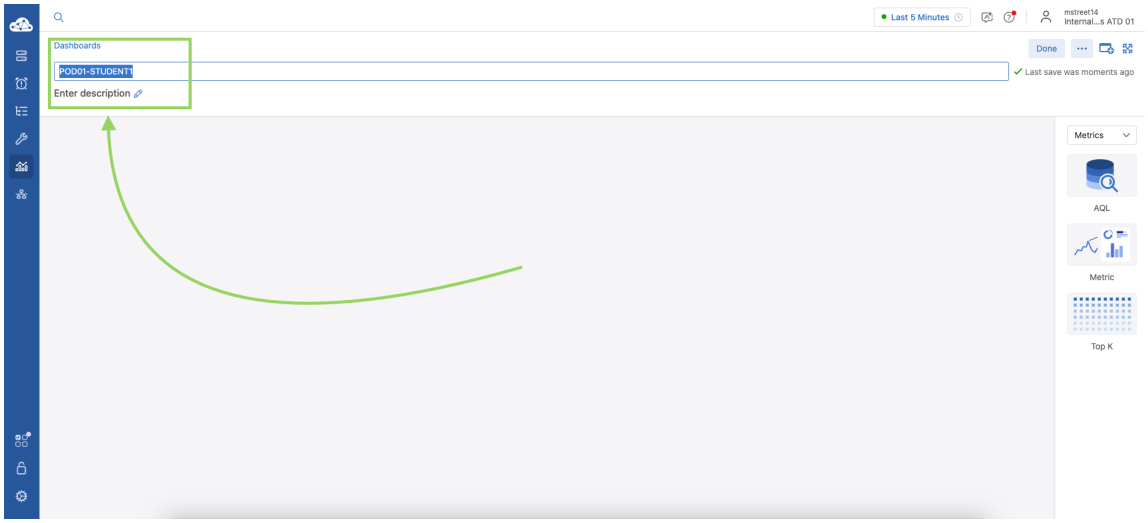
There is a lot to unpack in dashboards as you have a significant amount of power in customizing

the data and look of your dashboard. Let's walk through creating one step by step:

1. Select the `+ New Dashboard` button.



2. Provide a useful `Name` for the Dashboard, such as `POD01-STUDENT1`.



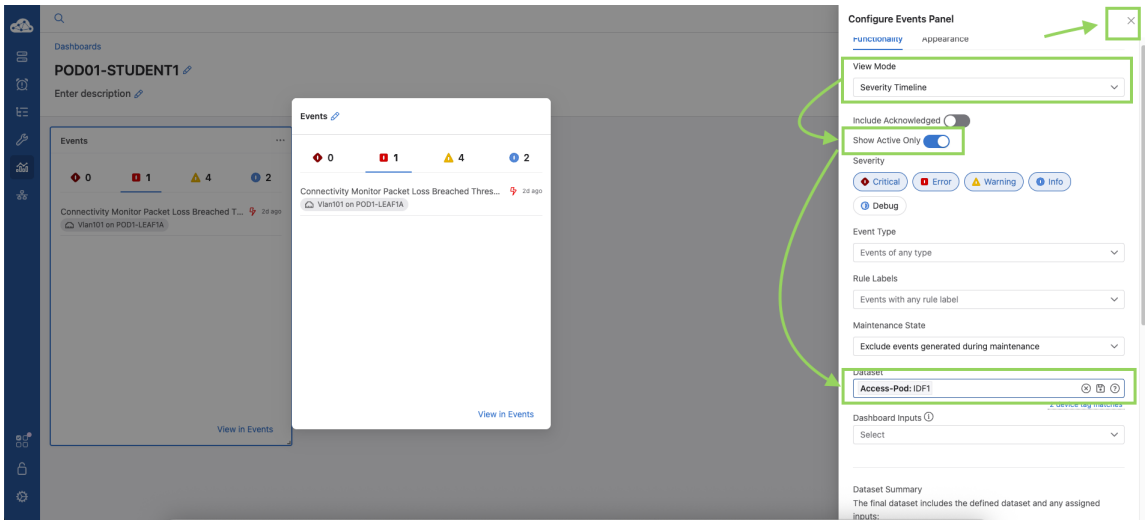
3. Next, let's add a combination of visualizations that we want to capture.
4. First, select the drop down on the upper right and change from `Metrics` to `Summaries`.
5. Within the `Summaries` list, select or drag the `Events` widget into the dashboard canvas.
6. Within the `Events` tile now added to your dashboard, select `Configure` from the three-dots ••• menu.



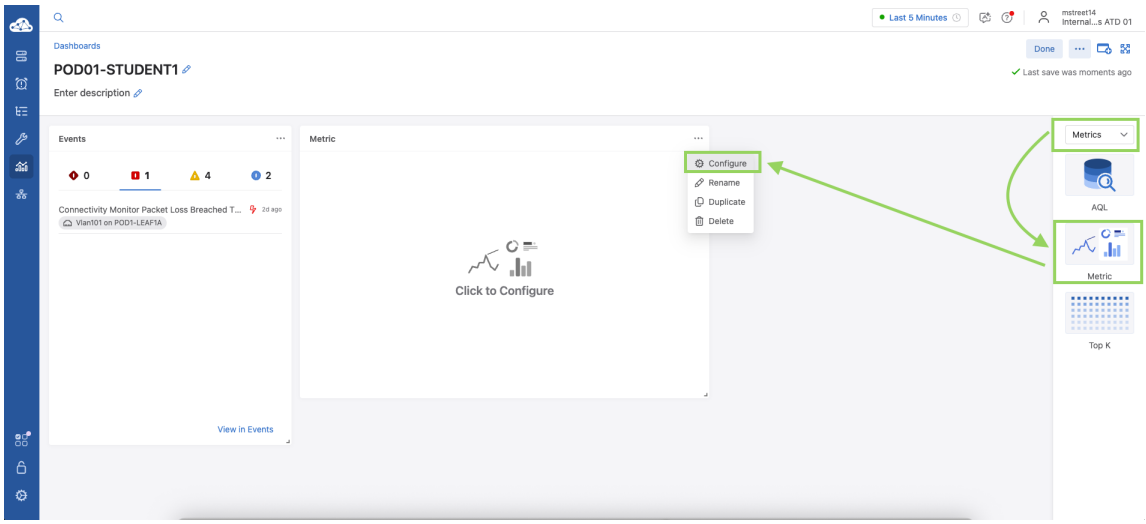
7. Within the right side menu bar, Student 1 is `IDF1` and Student 2 is `IDF2`. Select the following:

Events Settings

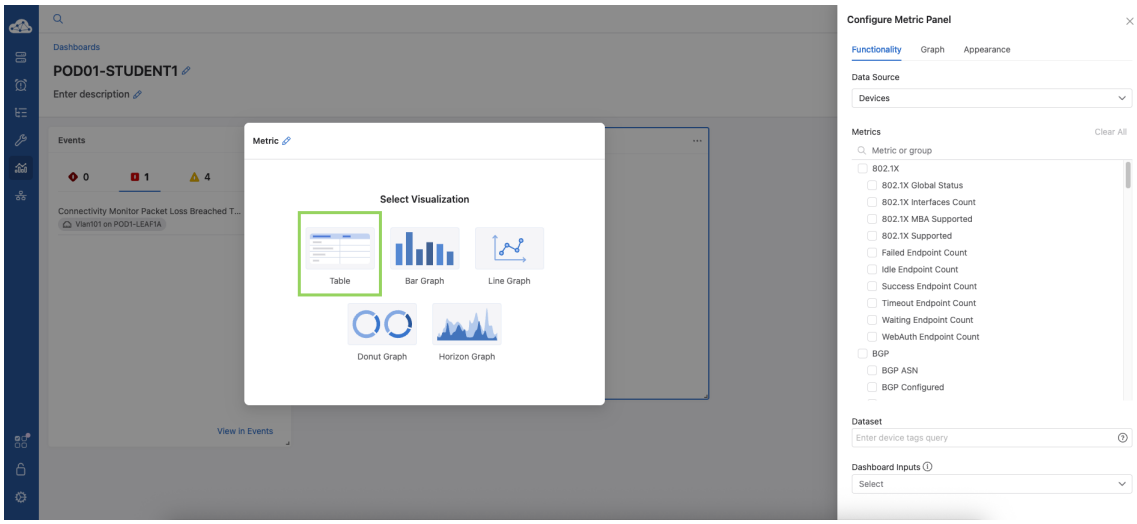
Key	Value
View Mode	Severity Timeline
Show Active Only	True
Dataset	(Student 1) `Access-Pod: IDF1`



8. Dismiss the customization menu by selecting the ✕ in upper right corner.
9. Next, change the `Summaries` menu back to `Metrics`.
10. Within the `Metrics` menu, select `Metric`. Then select `Configure` within the Metric tile.



11. Select `Table` from the `Select Visualization` menu.



12. Configure the `Table` using the `Configure Metric Panel` with the settings below:

Table Settings

Key	Value
Data Source	Devices
Dataset	(Student 1) `Access-Pod: IDF1`
Metrics #1	802.1X Interface Count
Metrics #2	CPU Utilization
Metrics #3	Total Power Granted
Metrics #4	ARP Table Size
Metrics #5	Boot Time

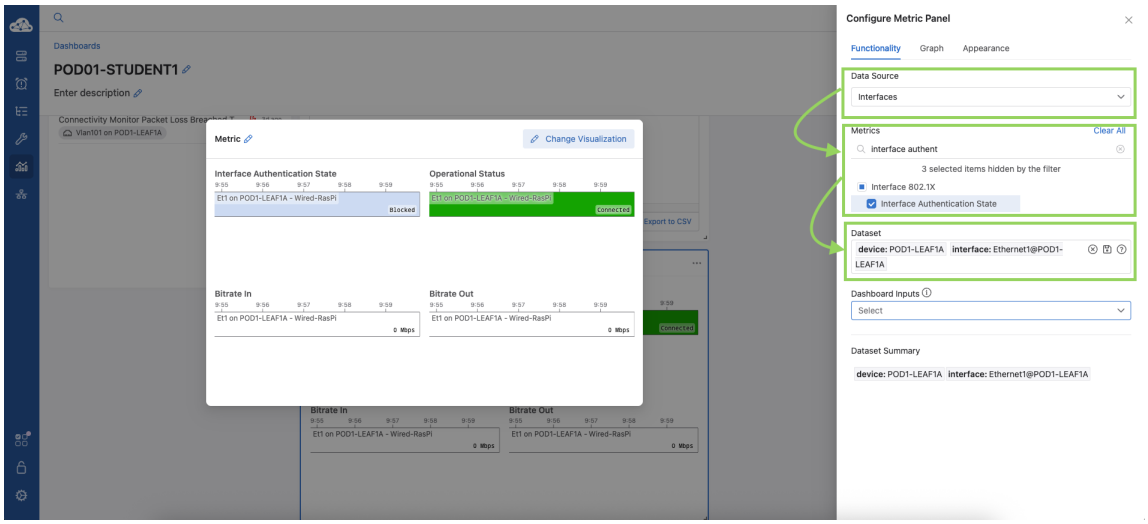


13. Dismiss the customization menu by selecting the ✕ button in upper right.
14. Follow the same steps to create a `Horizon Graph` using the information from the table below:

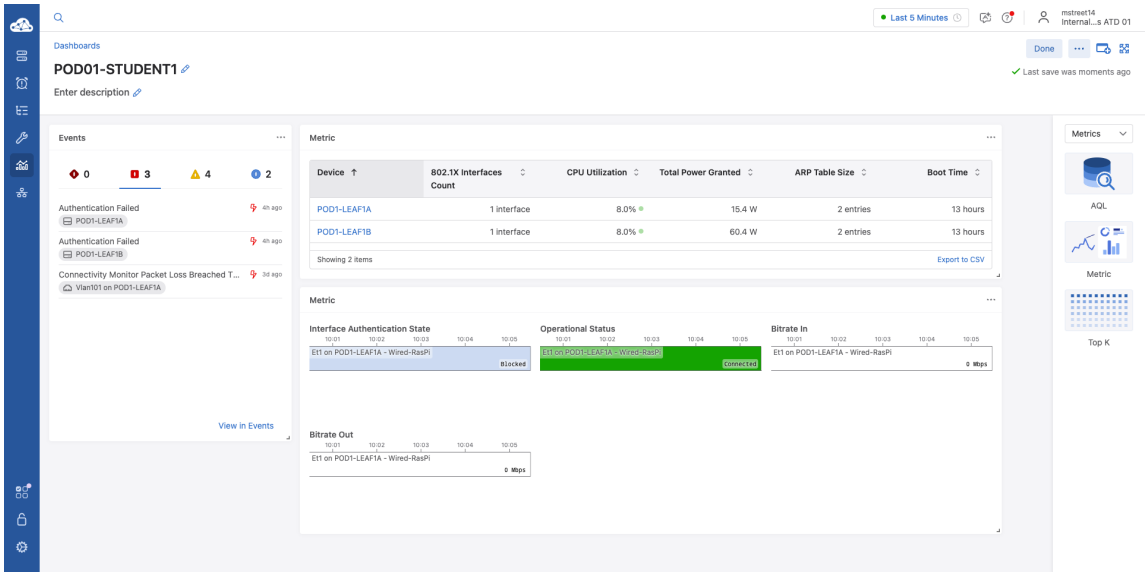
Horizon Graph Settings

Target your student device below!

Key	Value
Data Source	Interfaces
Dataset	(Student 1) `device: pod##-leaf1a`
Dataset	`interface: Ethernet1`
Metrics #1	Bitrate In
Metrics #2	Bitrate Out
Metrics #3	Operational Status
Metrics #4	Interface Authentication State



15. Dismiss the customization menu and resize the tiles as desired. You should have a new dashboard that looks something like this.

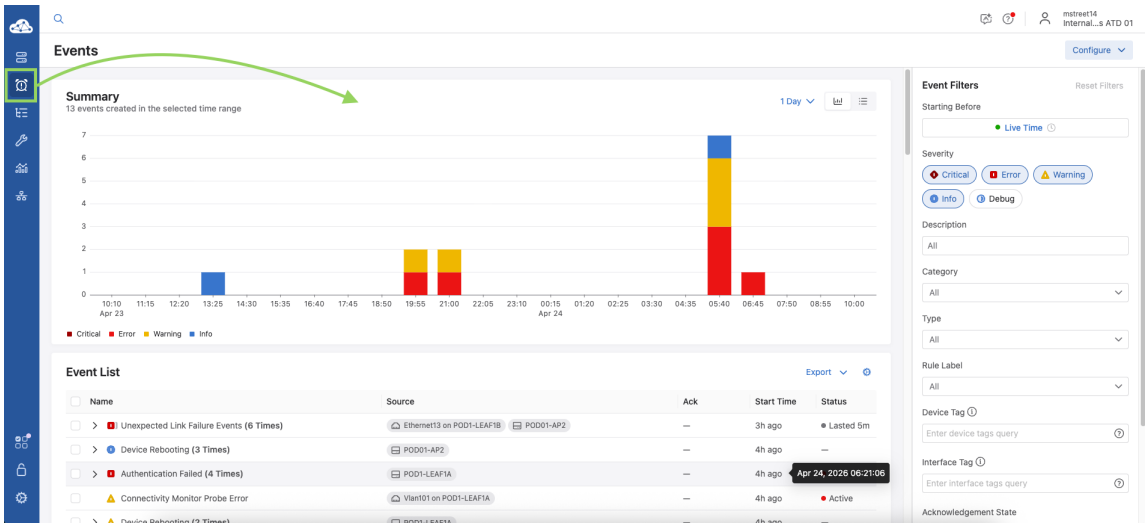


16. Save and complete the dashboard customization by selecting the `Done` button in the upper menu bar.

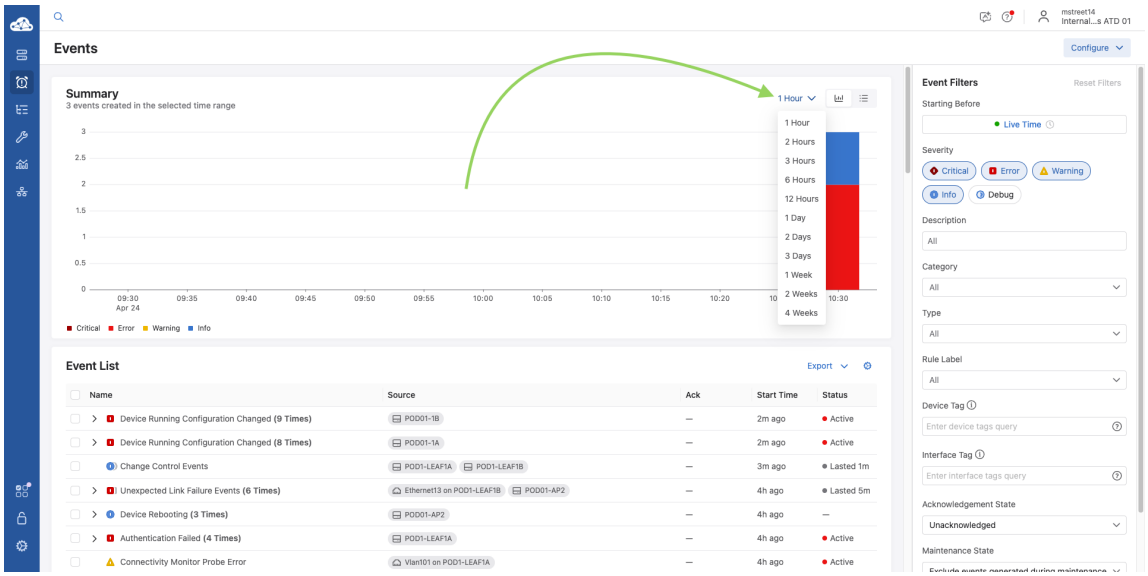
Events

In this section, we'll explore the Events stream and the tools and filters that help you process and manage critical errors versus informational data.

1. Open `Events` from the menu bar.

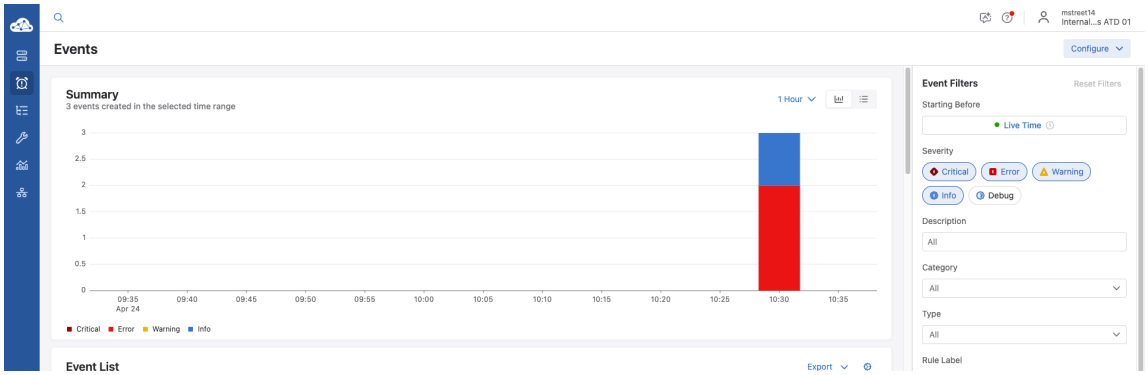


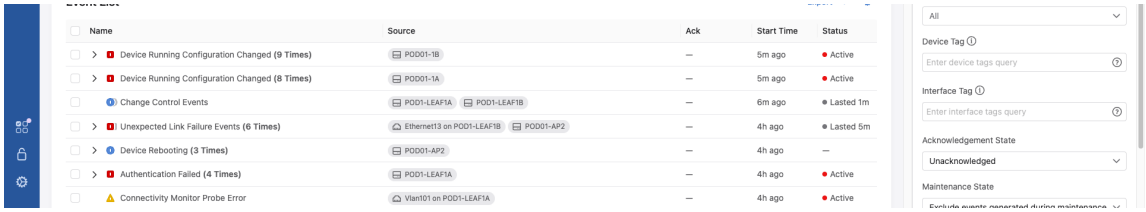
2. Select the current time selection and change it to `1 Hour` to adjust the summary visualization.



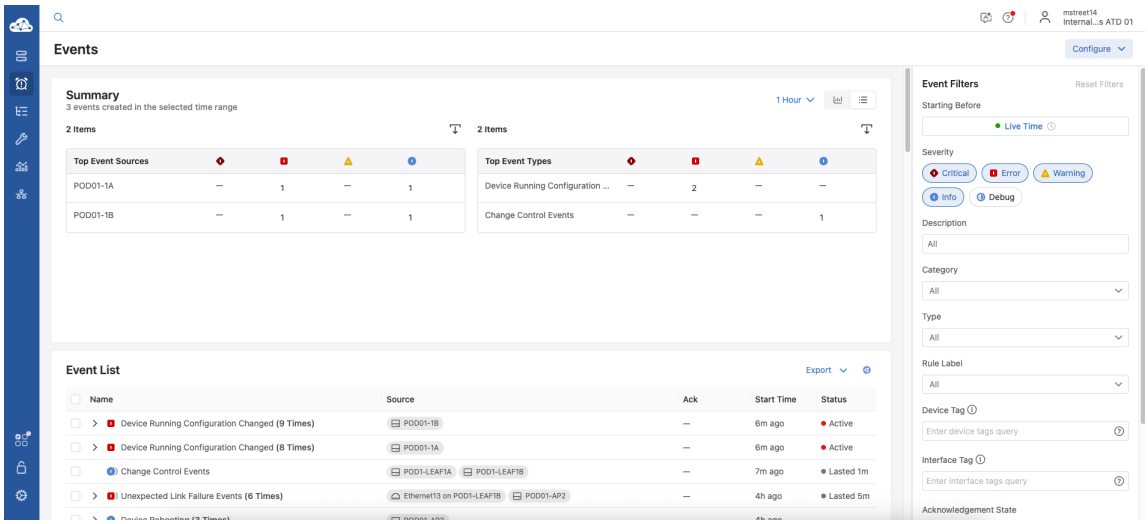
You can also toggle between an `Events Chart` and `Events Summary` table.

Events Chart

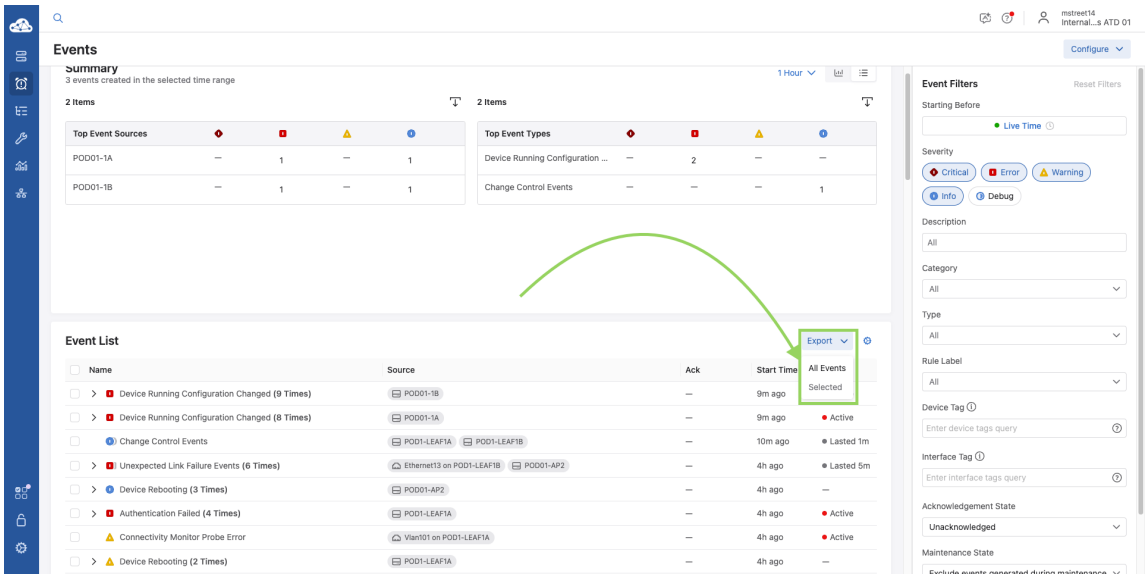




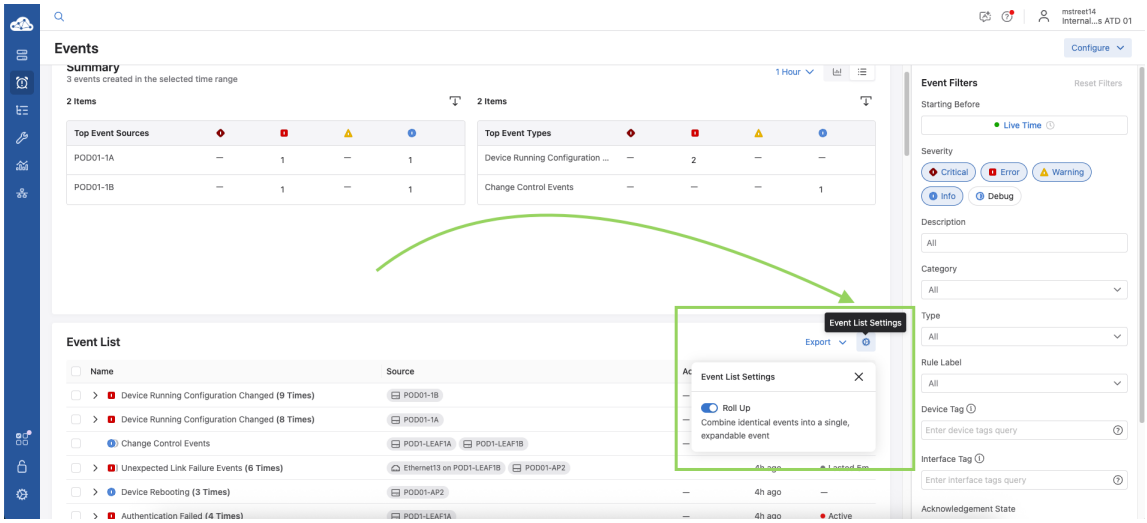
Events Summary



3. In the `Event List`, note the `Export` button to download the current event list as CSV. You can download `All Events` or only `Selected`.



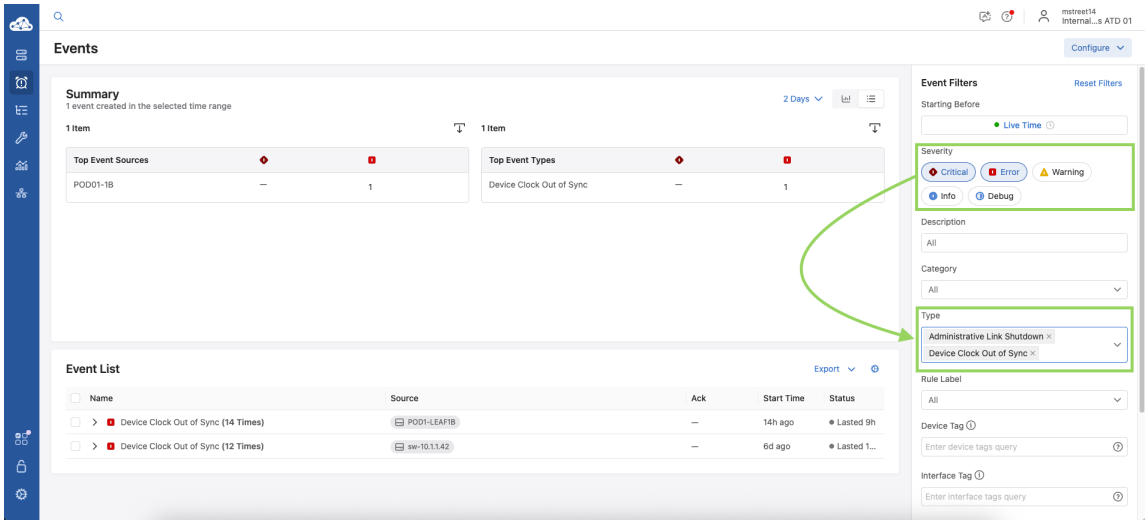
4. Select the Gear ⚙ icon to open `Event List Settings` and toggle `Roll Up`. This setting combines repeated events into groups. Toggle this on and off to see the effect on the list of events.



5. Apply the Event Filters on the right pane to reduce the amount of data displayed. One student should do this portion.

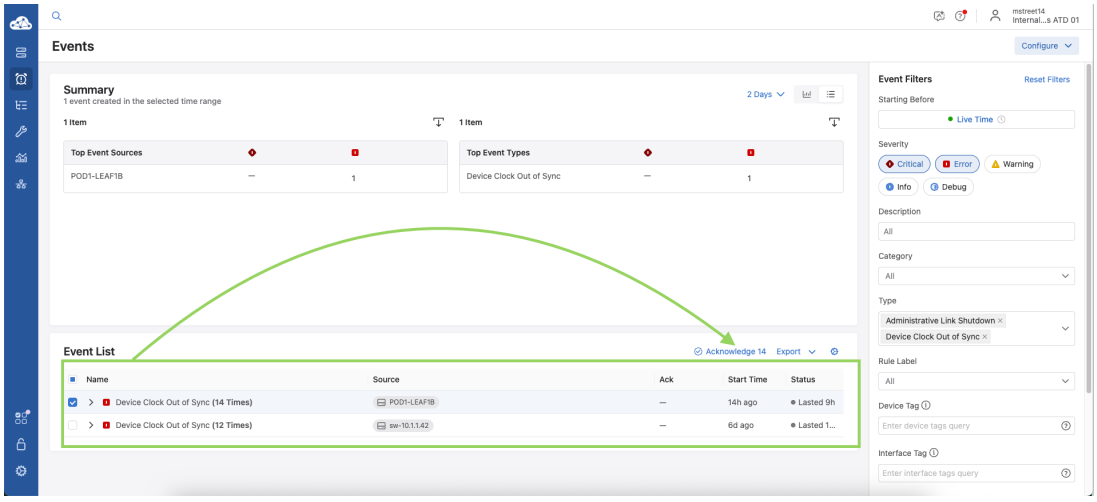
Filter Settings

Key	Value
Severity	`Critical`, `Error`
Type #1	`Administrative Link Shutdown`
Type #2	`Device Clock Out of Sync`

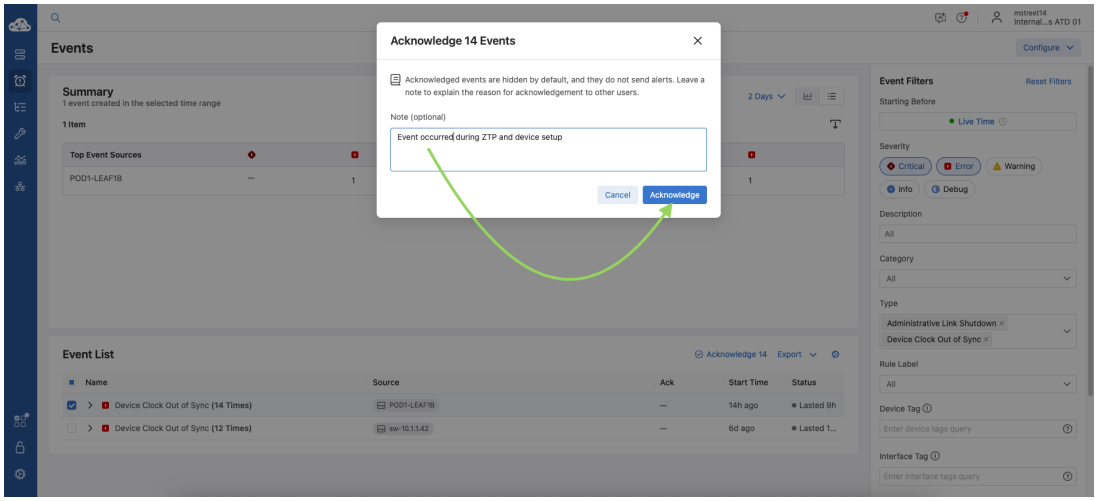


6. Acknowledge and unacknowledge events:

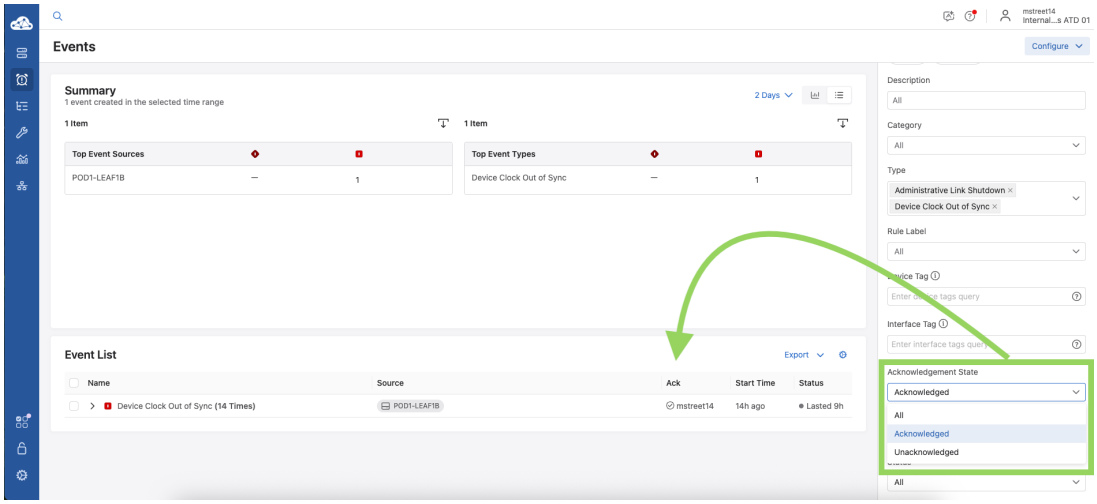
   a. From the filtered event list, select specific events and select `Acknowledge`.



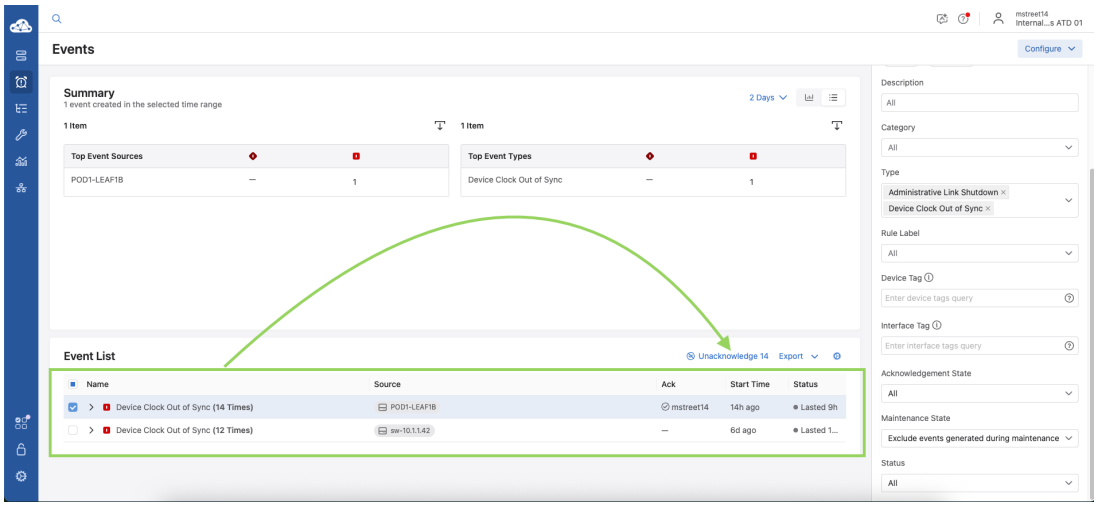
b. Adding a note is optional. Select `Acknowledge` to mark these selected events.



c. Acknowledged events are not deleted — they're flagged and can be referenced by changing the `Acknowledgement State` filter. Select `Acknowledgement State` and select `Acknowledged`.



d. To unacknowledge an event, select the box to the left of the acknowledged event and select `Unacknowledge` at the top of the event list.

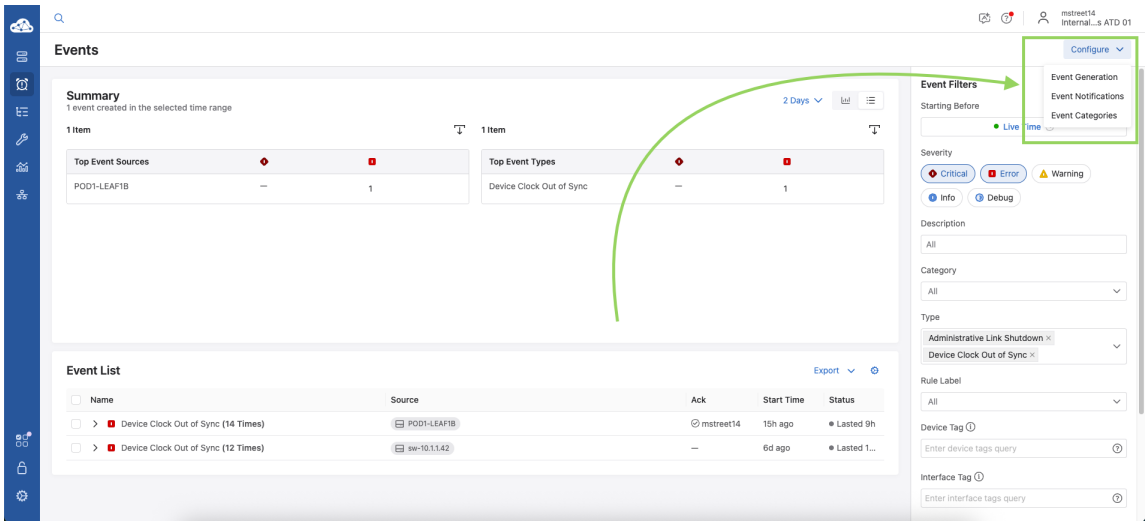


7. Events and filtering section complete!

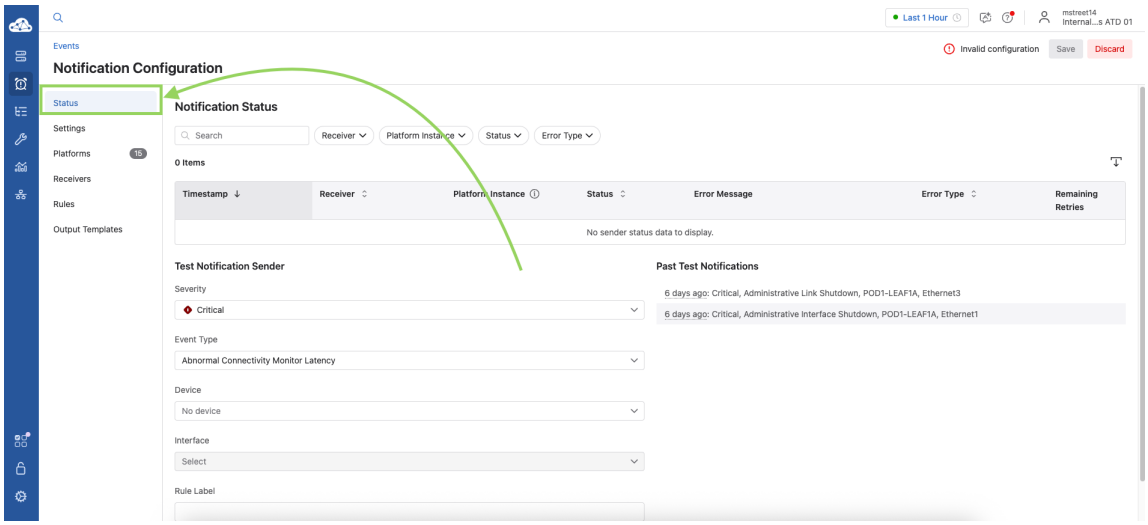
Customize Notifications

In this section, you'll configure CloudVision to send an email alert using the built-in SendGrid email service. CloudVision natively supports other notification systems as well (chat, SNMP, Syslog, etc.), but we'll focus on email for this lab.

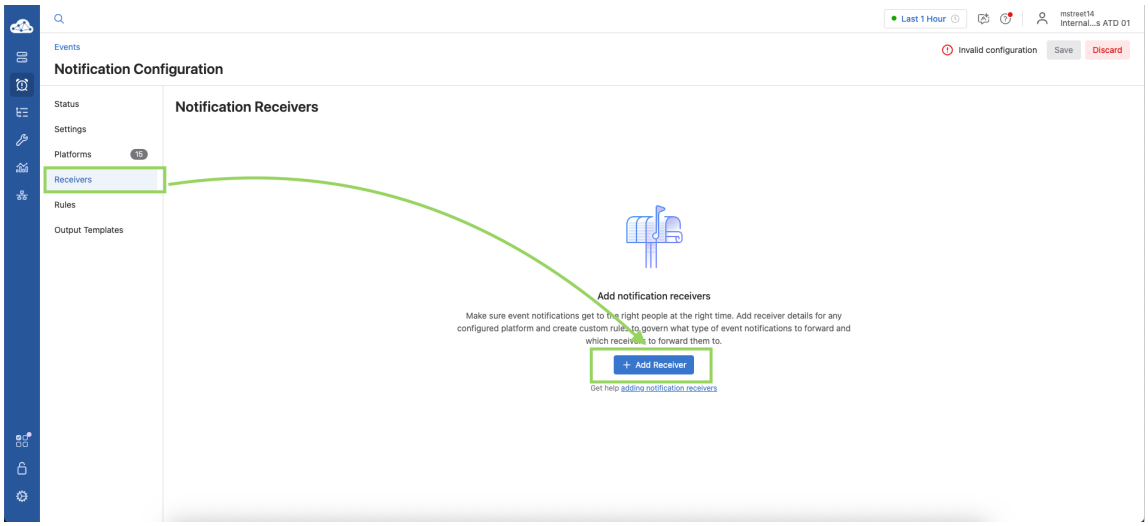
1. If you are not already on the `Events` page, select it from the menu.
2. Select the `Configure` dropdown and select `Event Notifications`.



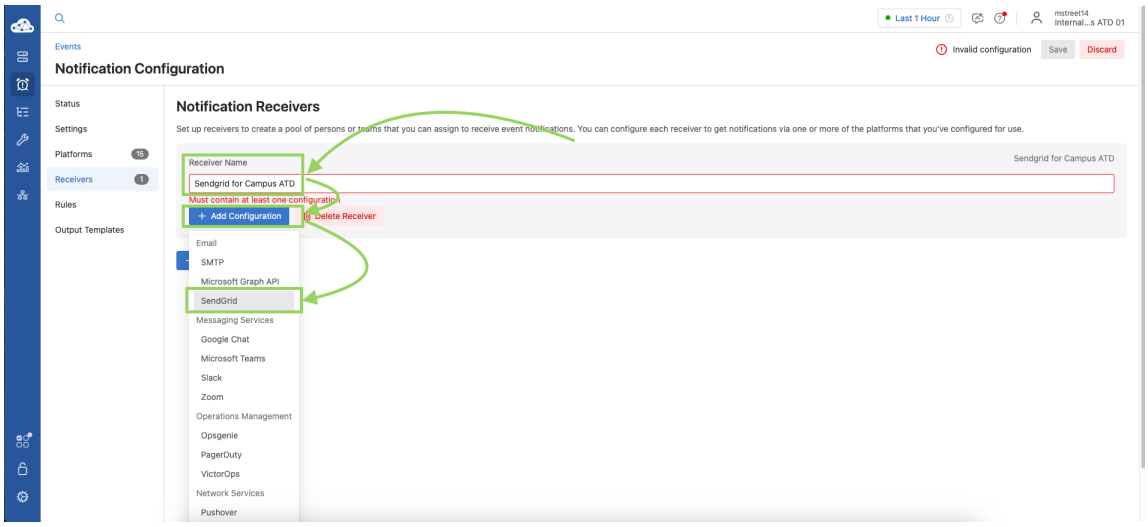
3. The notification status page shows 0 items configured.



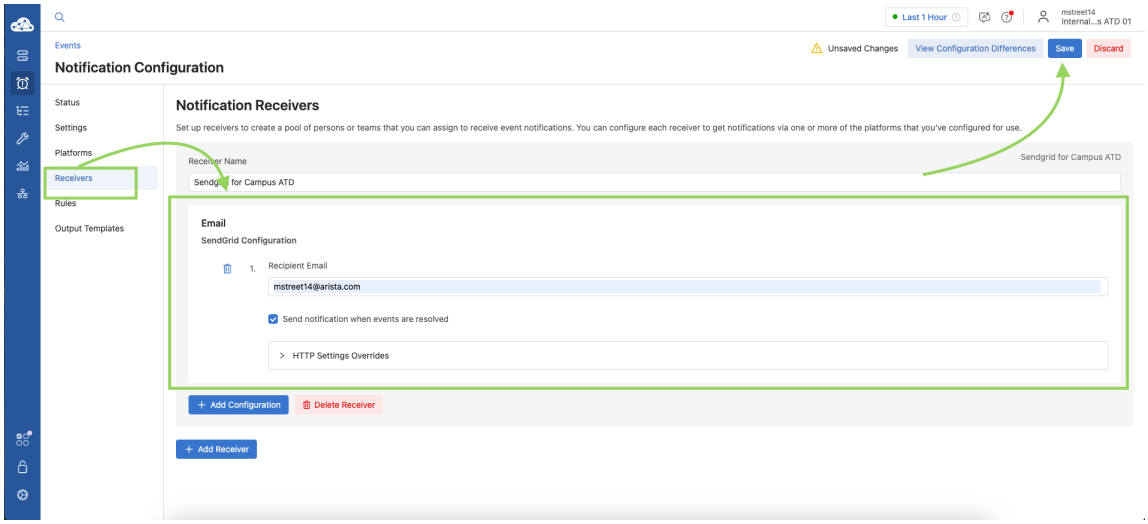
4. Navigate to `Receivers` in the menu, then select `Add Receiver`.



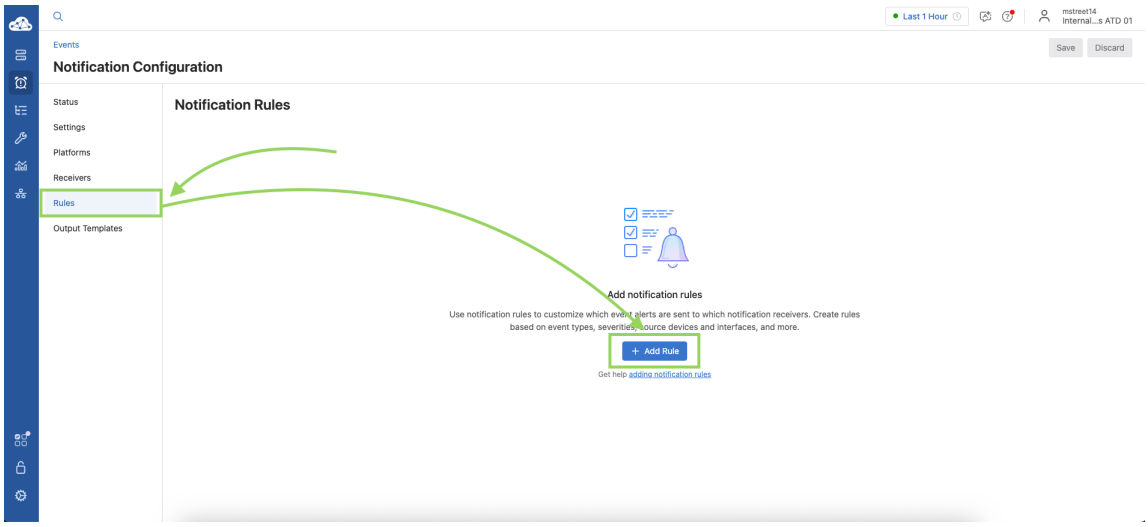
5. Name the receiver `SendGrid for Campus ATD`, then select `Add Configuration` and select `SendGrid` from the menu options.



6. Enter a valid email address that you can receive emails at during this lab and select the `Send notification when events are resolved` checkbox. Select `Save` in the upper right to save your new receiver.

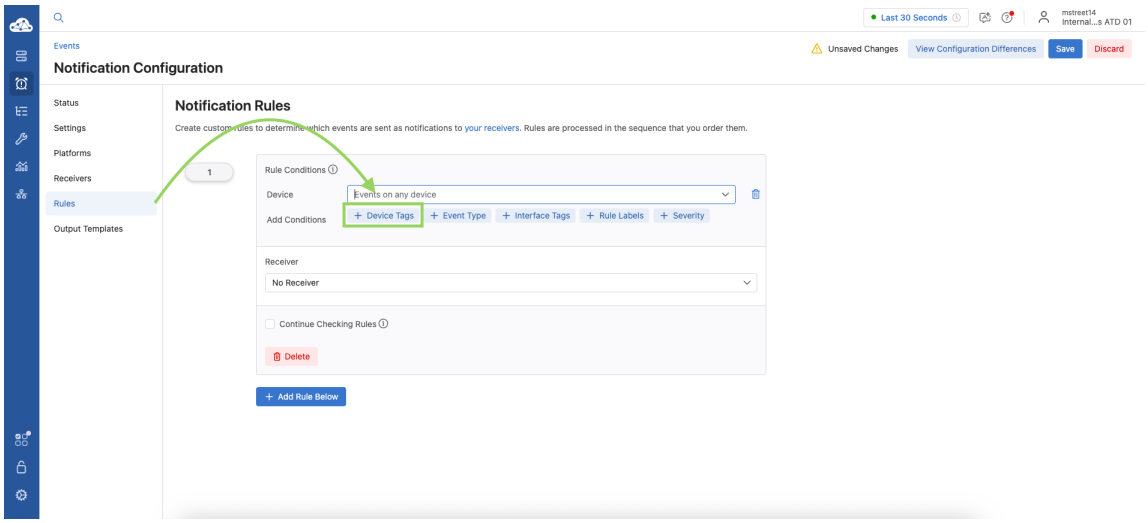


7. Once you are happy with the receiver's configuration, select `Save` at the top of the screen.
8. Next, let's configure a rule to use the new receiver. Navigate to `Rules`, then select `Add Rule`.

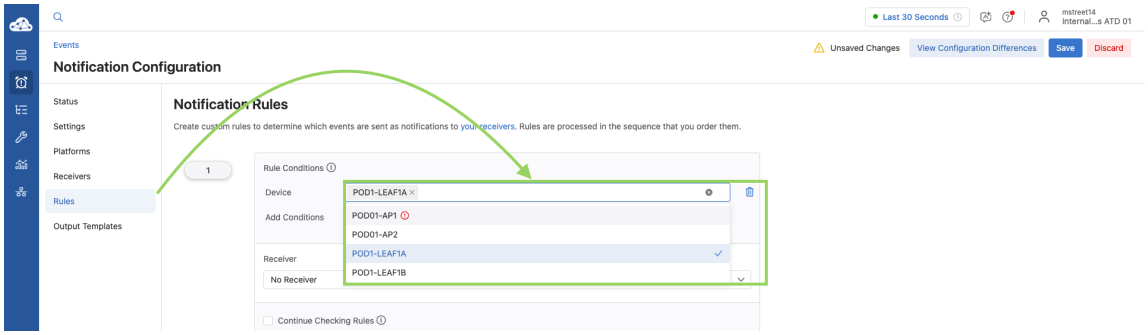


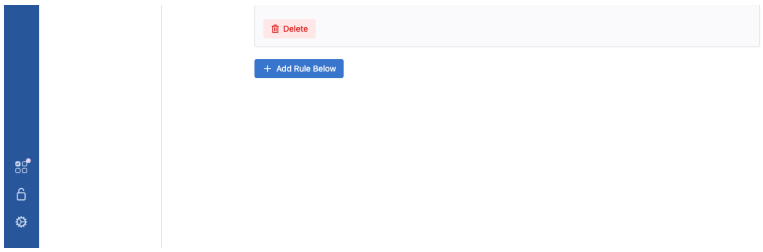
9. Configure `Rule Conditions` for this rule. Select `Device Tags`, then choose your switch from the `Device` dropdown.

Device Tags



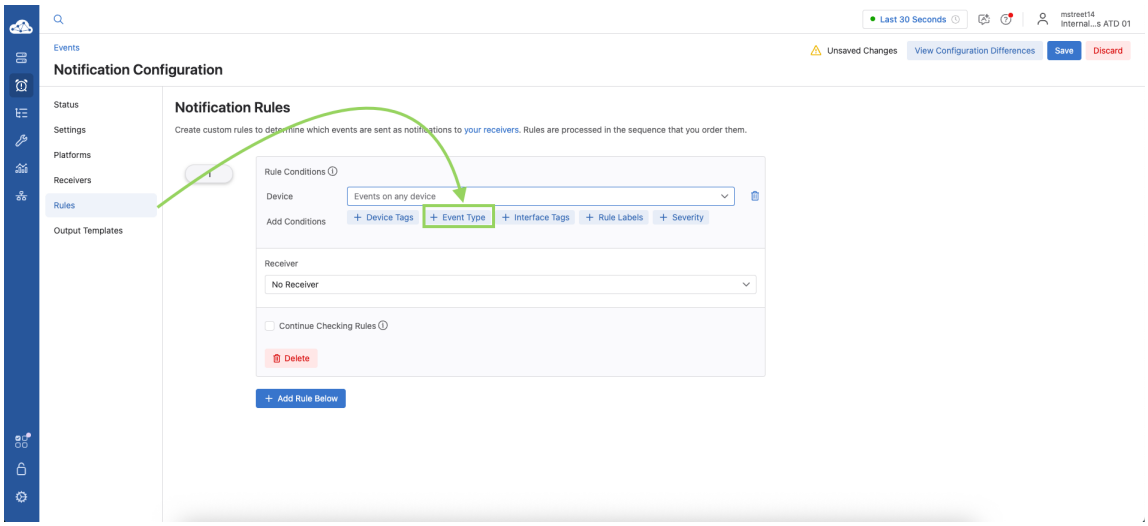
Select Device



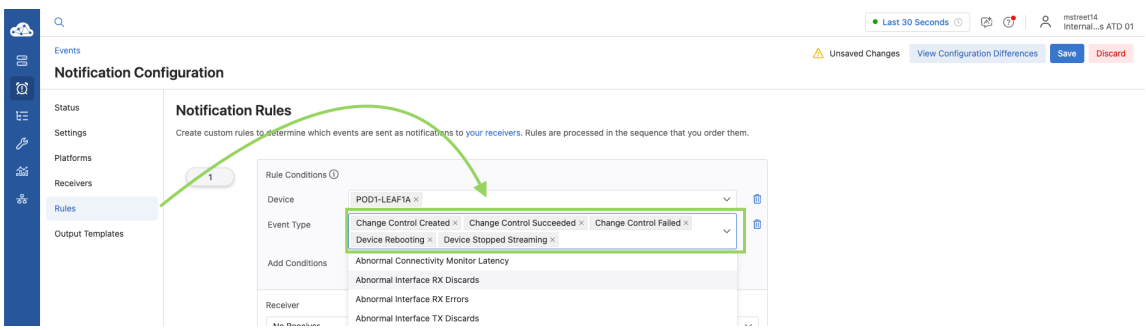


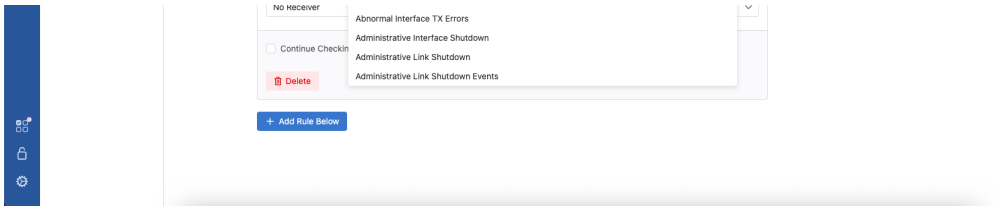
10. Select `+ Event Type`. Add event types by choosing them from the dropdown as shown in the image below:

Add Event Type

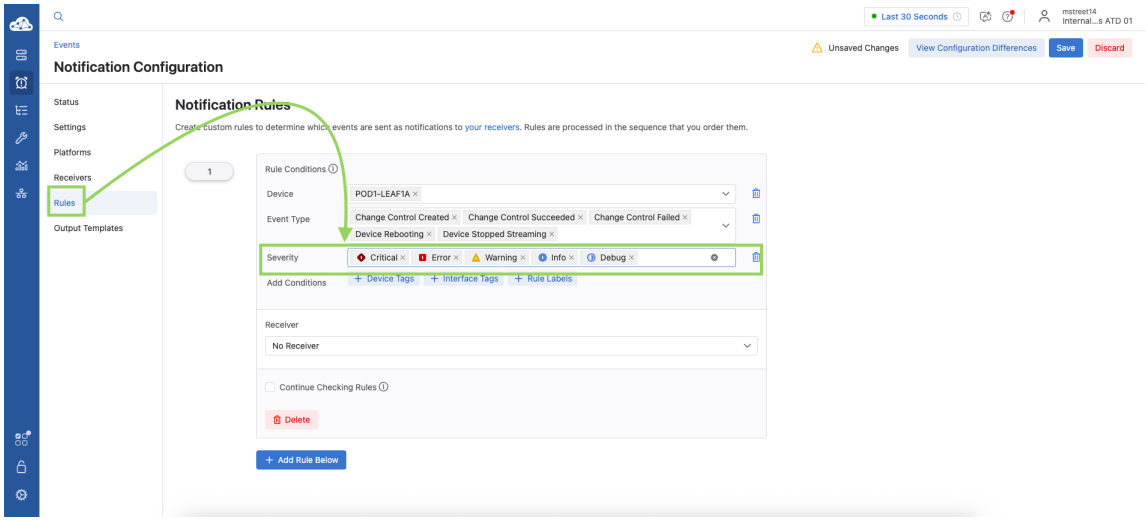


Select Event Types



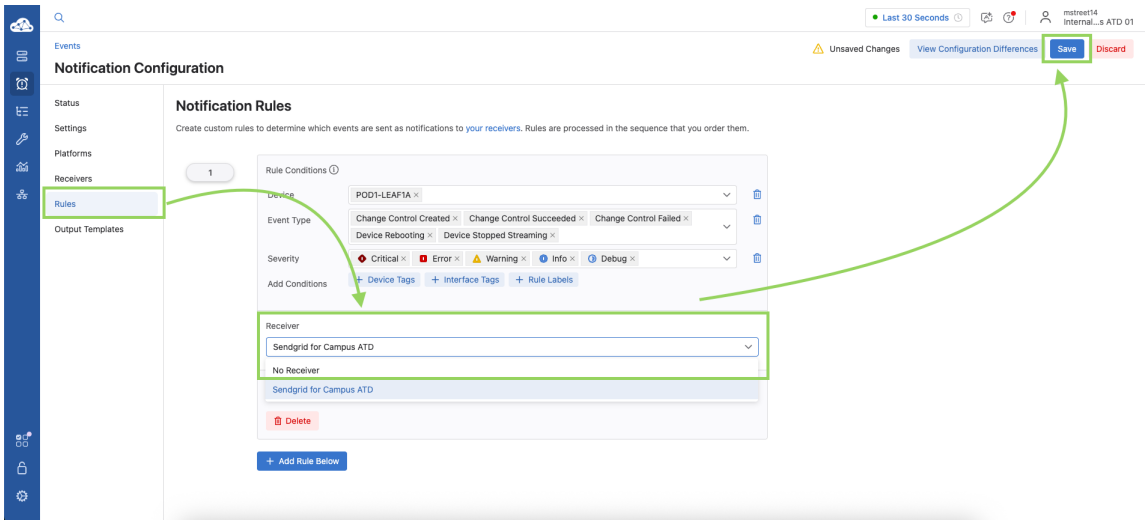


11. Select all severity options by selecting `+ Severity` and choosing the options from the dropdown.

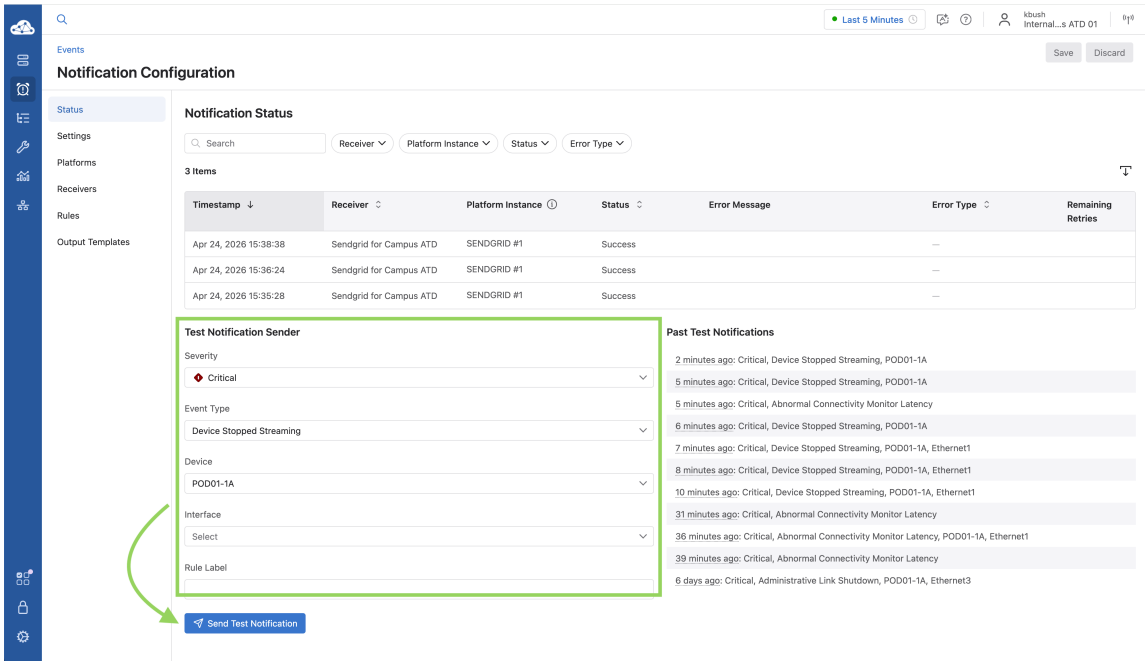


12. Select your `SendGrid for Campus ATD` receiver from the `Receiver` dropdown, select the `Continue Checking Rules` box, and select `Save`.

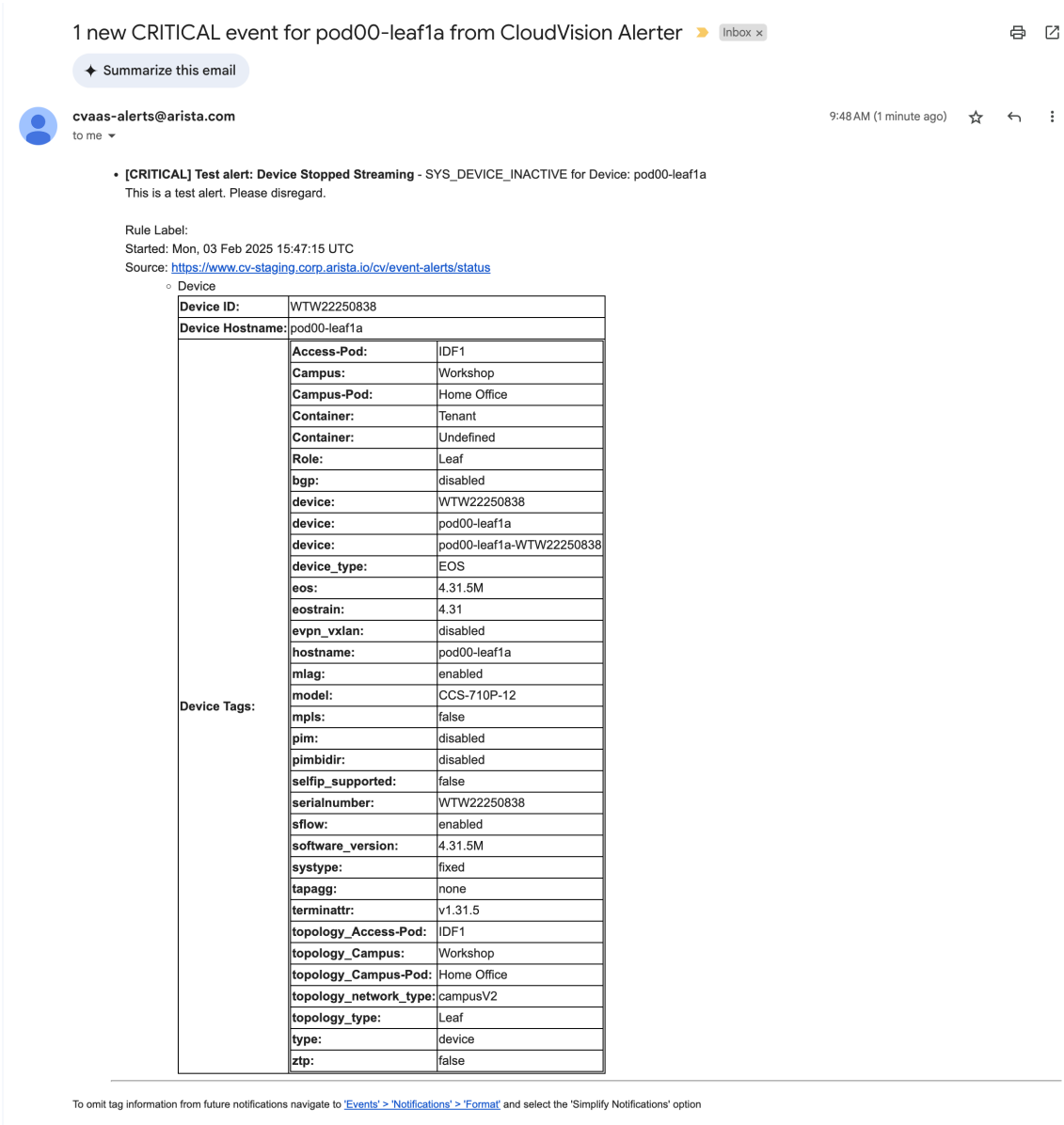
> **Save your changes!**
>
> Make sure to save your changes on this screen with the `Save` button along the top of your screen.



13. Now let's test your new receiver and rule. Navigate to `Status`. Configure the `Test Notification Sender` by changing the `Event Type` to `Device Stopped Streaming` and selecting your switch from the `Device` dropdown. Select `Send Test Notification`.



14. After a minute or two, you should receive the email alert at the email address you configured in the receiver.



1 new CRITICAL event for pod00-leaf1a from CloudVision Alerter Inbox ×

✦ Summarize this email

cvaas-alerts@arista.com
to me

9:48 AM (1 minute ago)

- **[CRITICAL] Test alert: Device Stopped Streaming** - SYS_DEVICE_INACTIVE for Device: pod00-leaf1a
  This is a test alert. Please disregard.

  Rule Label:
  Started: Mon, 03 Feb 2025 15:47:15 UTC
  Source: https://www.cv-staging.corp.arista.io/cv/event-alerts/status
  - Device

Device ID:	WTW22250838	
Device Hostname:	pod00-leaf1a	
Device Tags:	**Access-Pod:**	IDF1
	Campus:	Workshop
	Campus-Pod:	Home Office
	Container:	Tenant
	Container:	Undefined
	Role:	Leaf
	bgp:	disabled
	device:	WTW22250838
	device:	pod00-leaf1a
	device:	pod00-leaf1a-WTW22250838
	device_type:	EOS
	eos:	4.31.5M
	eostrain:	4.31
	evpn_vxlan:	disabled
	hostname:	pod00-leaf1a
	mlag:	enabled
	model:	CCS-710P-12
	mpls:	false
	pim:	disabled
	pimbidir:	disabled
	selfip_supported:	false
	serialnumber:	WTW22250838
	sflow:	enabled
	software_version:	4.31.5M
	systype:	fixed
	tapagg:	none
	terminattr:	v1.31.5
	topology_Access-Pod:	IDF1
	topology_Campus:	Workshop
	topology_Campus-Pod:	Home Office
	topology_network_type:	campusV2
	topology_type:	Leaf
	type:	device
	ztp:	false

To omit tag information from future notifications navigate to 'Events' > 'Notifications' > 'Format' and select the 'Simplify Notifications' option

15. You can remove your event receiver to avoid additional emails before continuing!

Additional References

- Interface Diagnostics
- AGNI TACACS

🎉 CONGRATS! You have completed the Wired labs! 🎉

2026-06-10